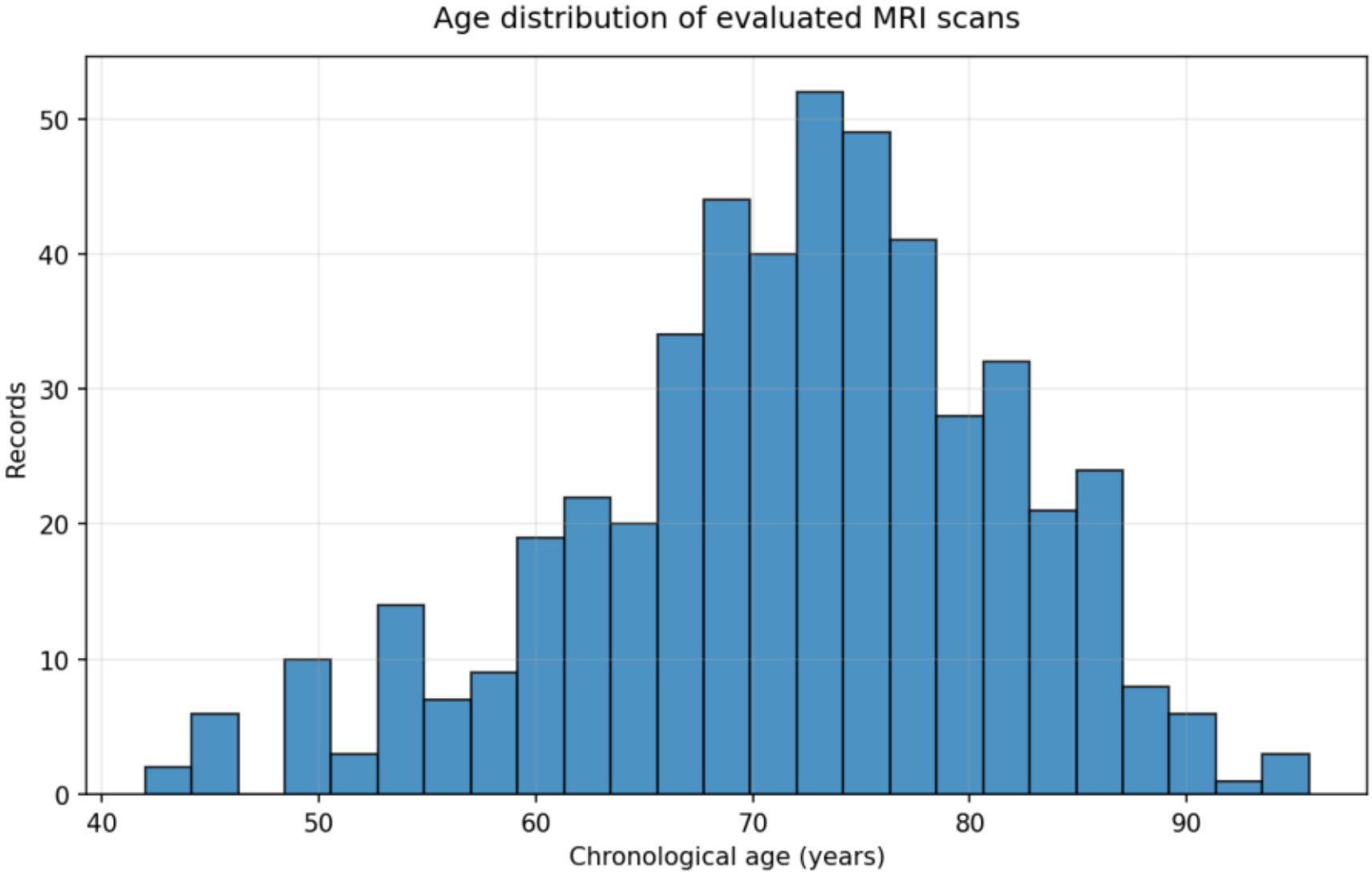
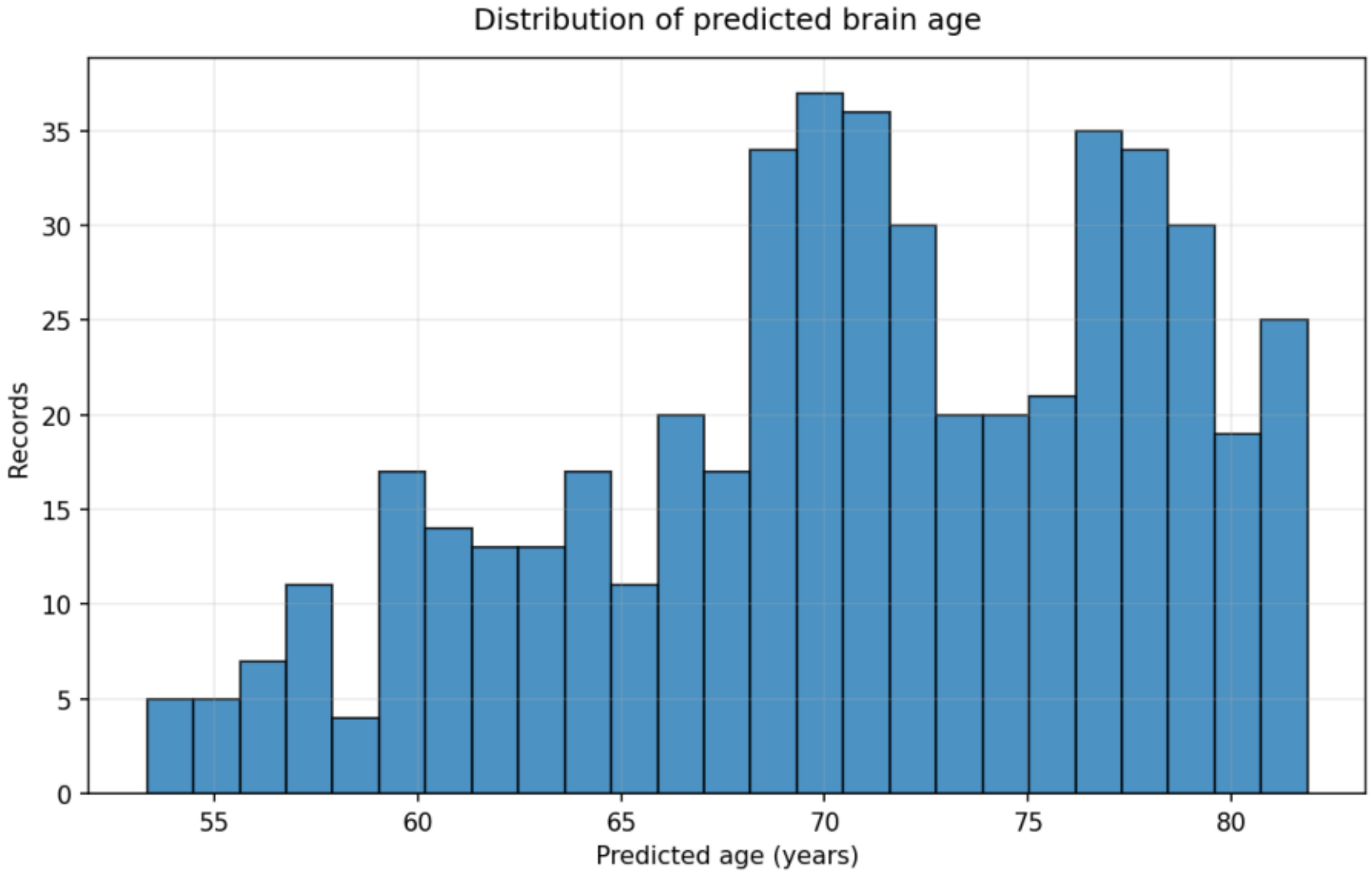
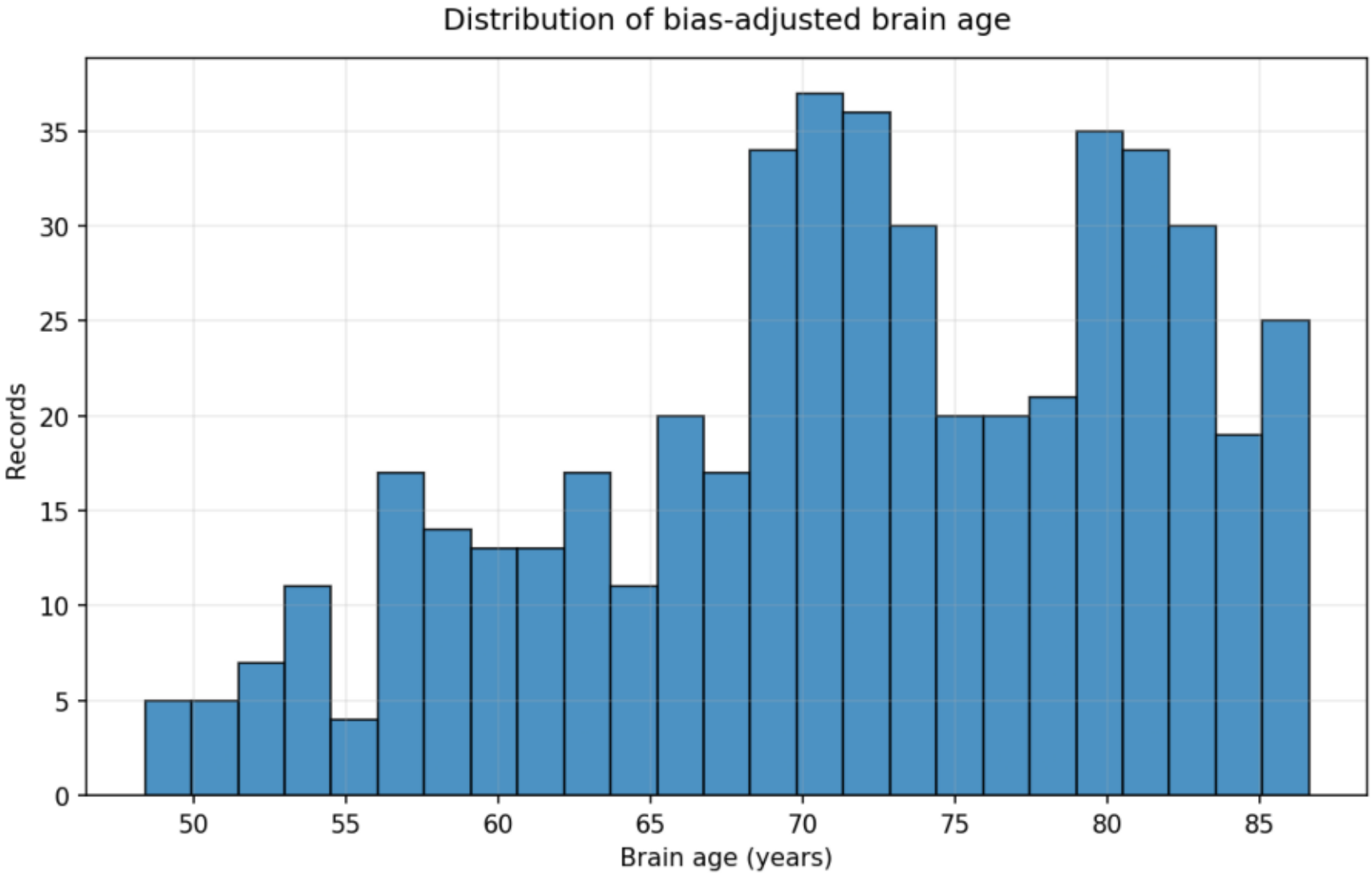
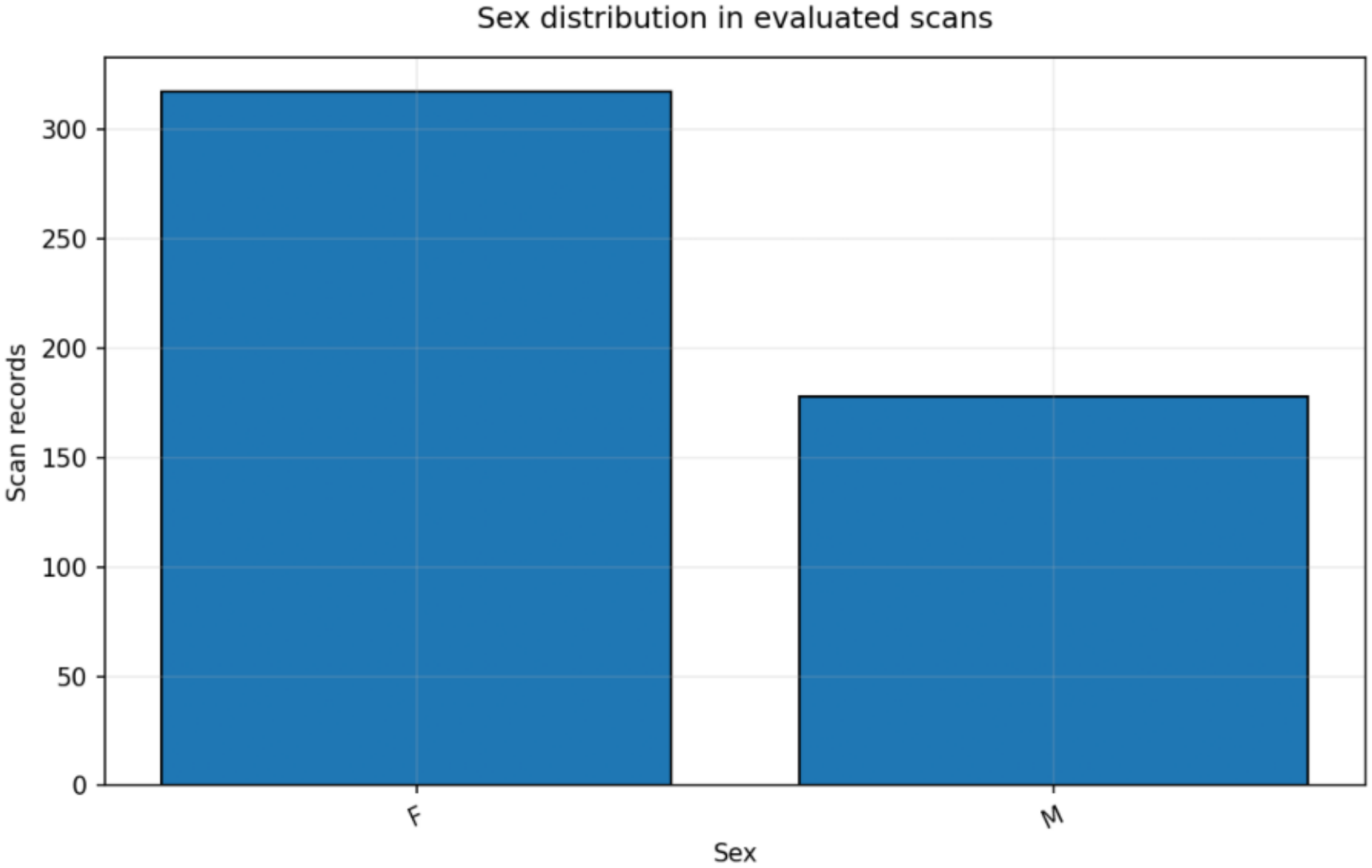
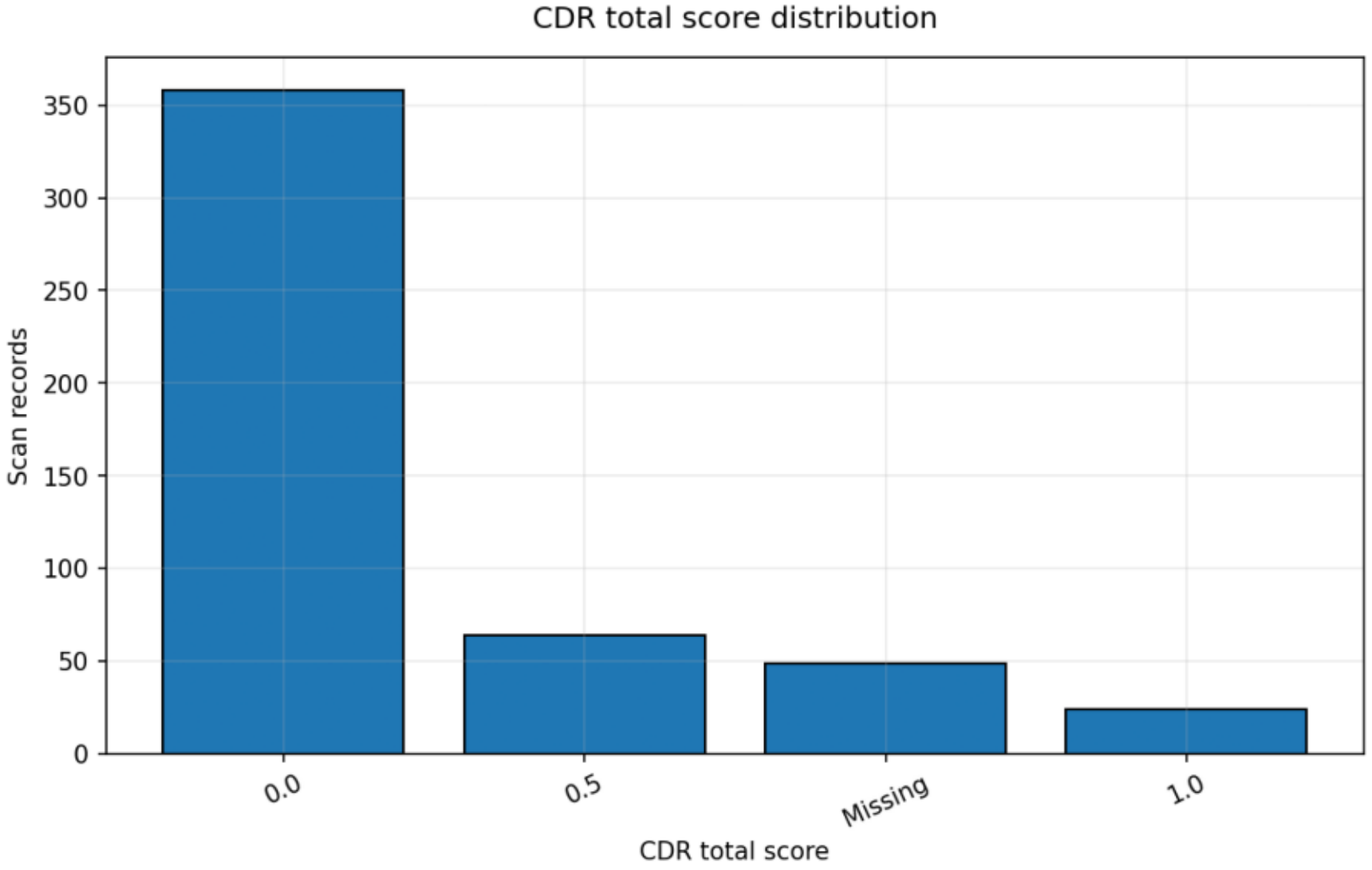
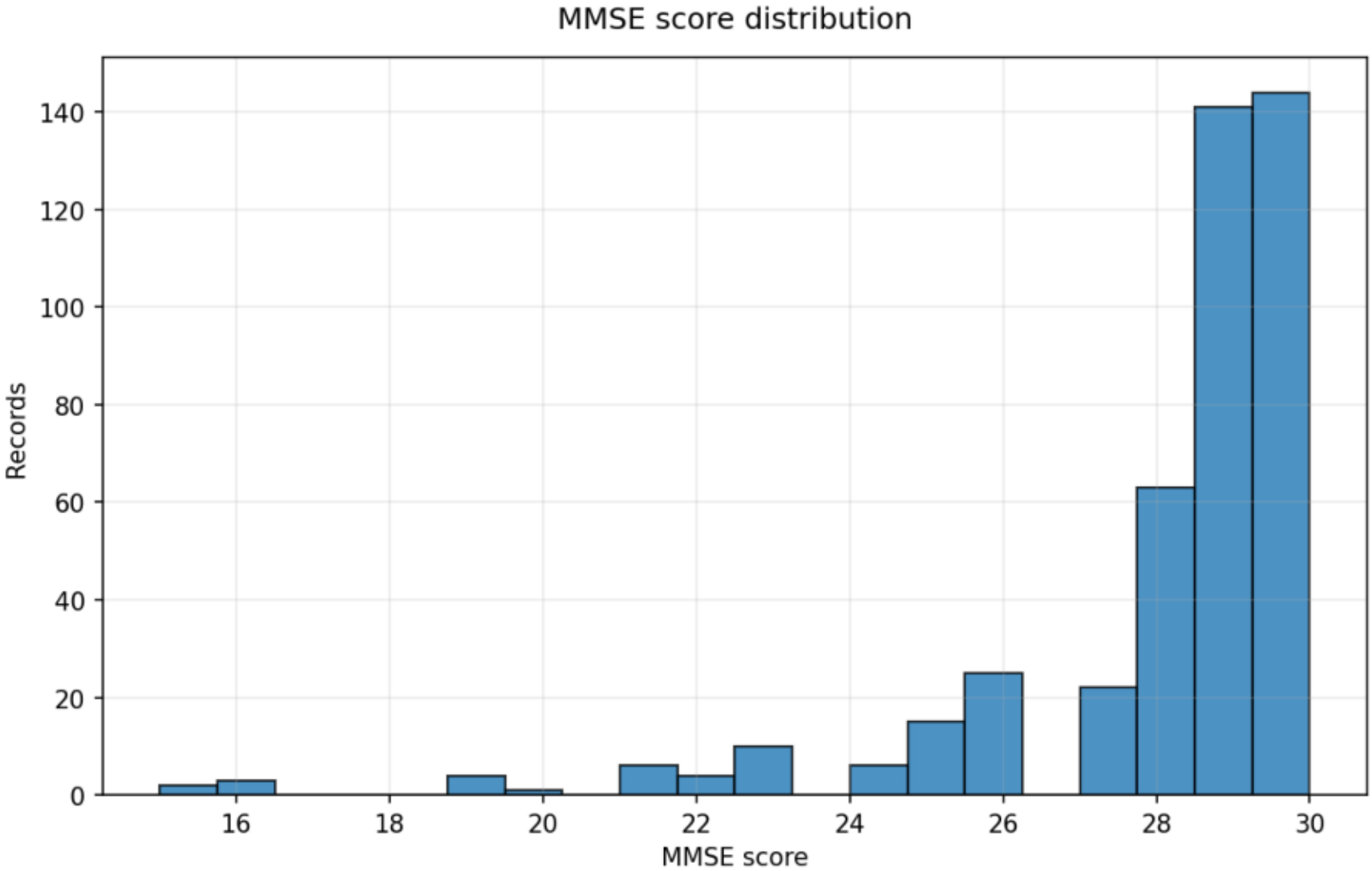

Age distribution of evaluated MRI scans
Records
0
10
20
30
40
50
40
50
60
70
80
90
Chronological age (years)

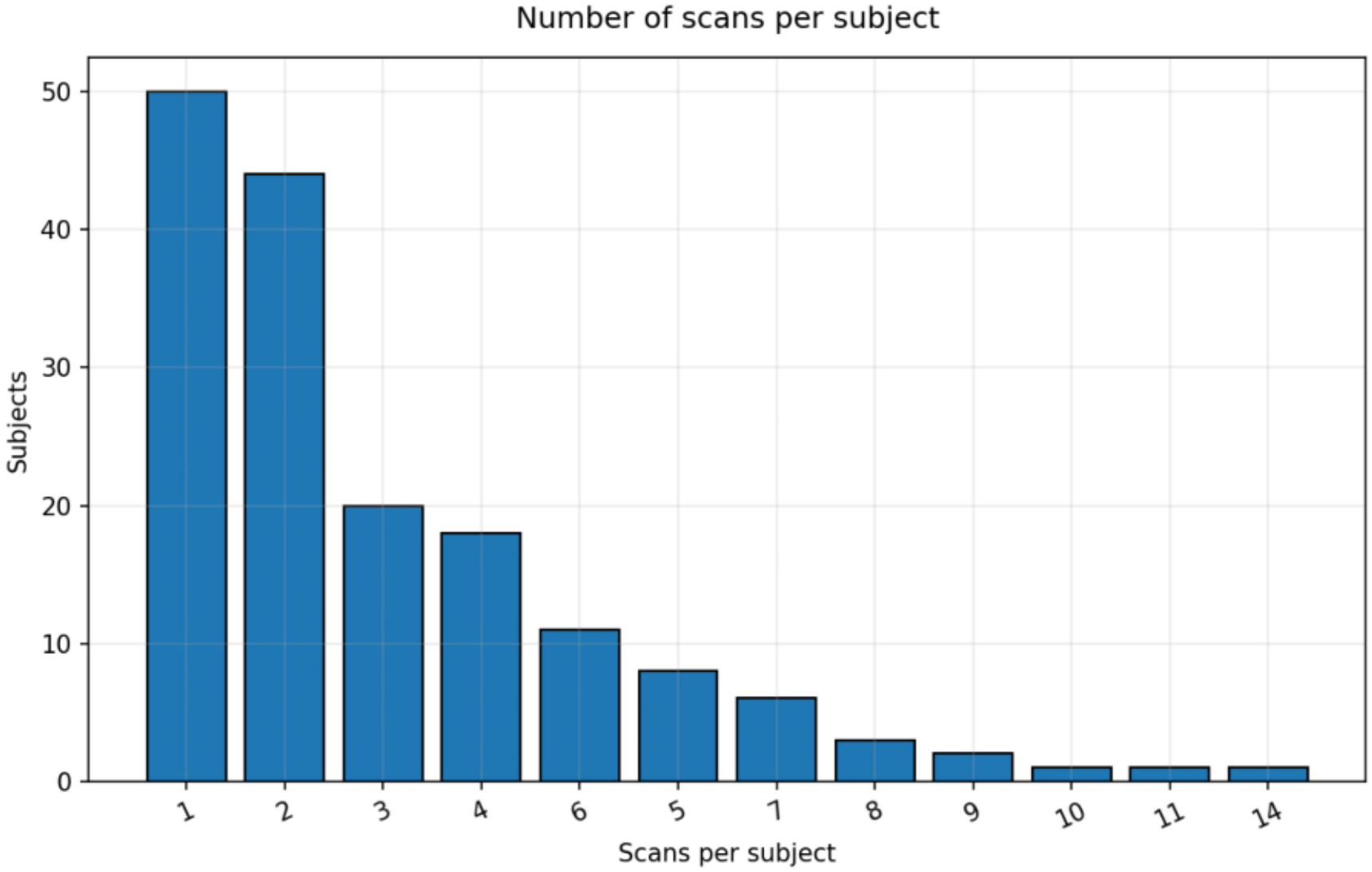

Distribution of predicted brain age
Records
0
5
10
15
20
25
30
35
55
60
65
70
75
80
Predicted age (years)

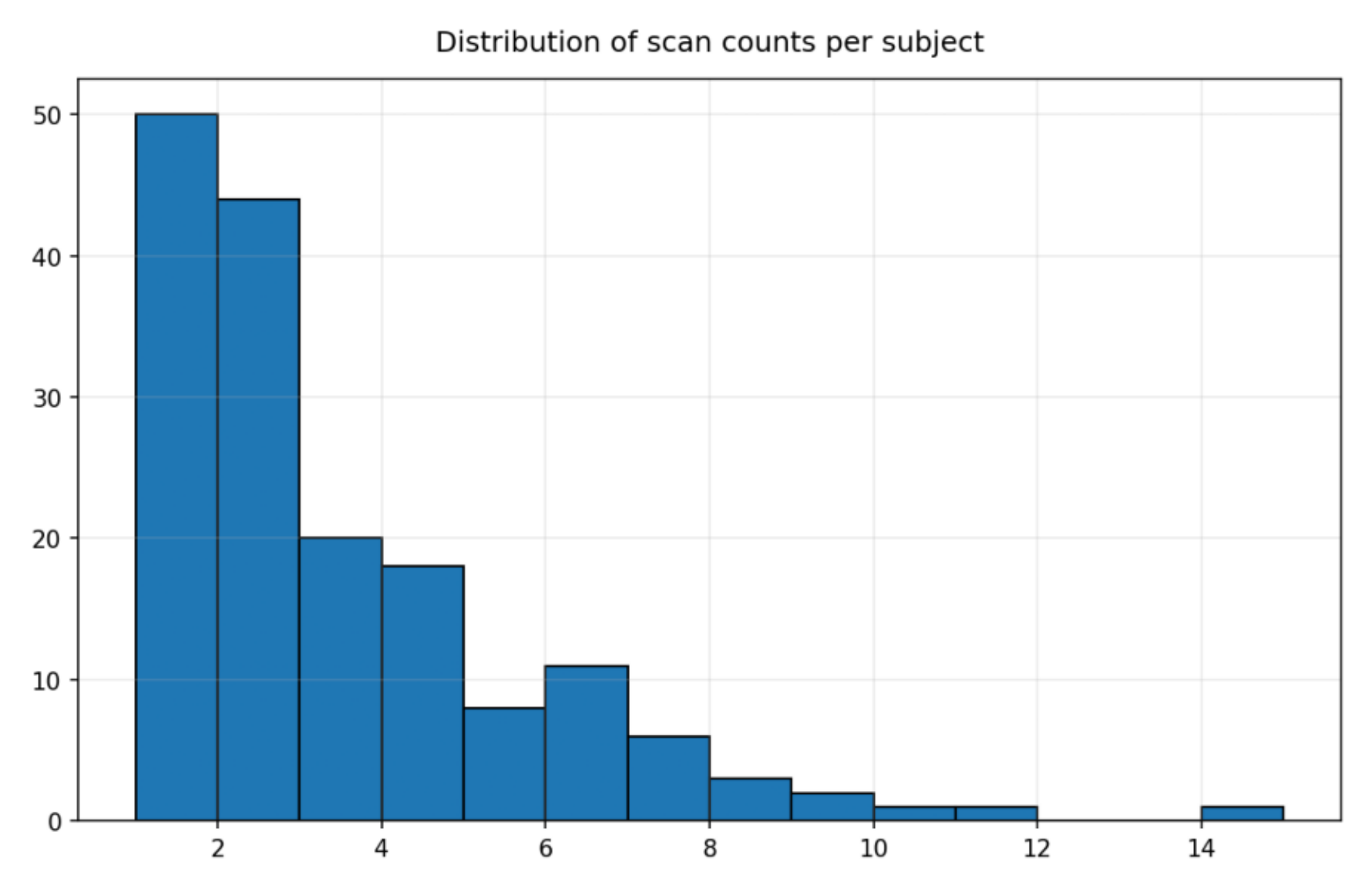

Distribution of bias-adjusted brain age
Records
0
5
10
15
20
25
30
35
50
55
60
65
70
75
80
85
Brain age (years)

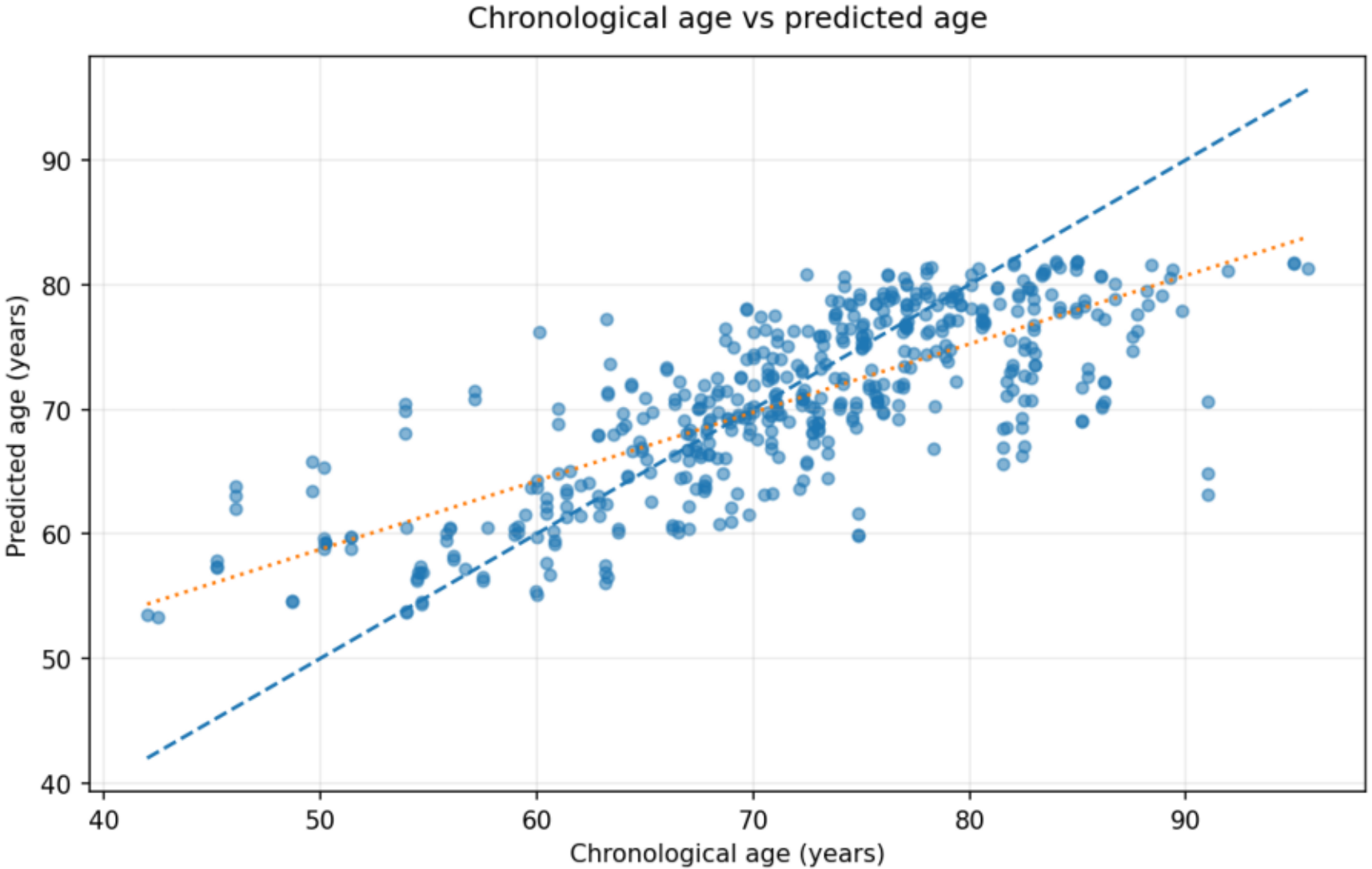

Sex distribution in evaluated scans
Scan records
0
50
100
150
200
250
300
F
M
Sex

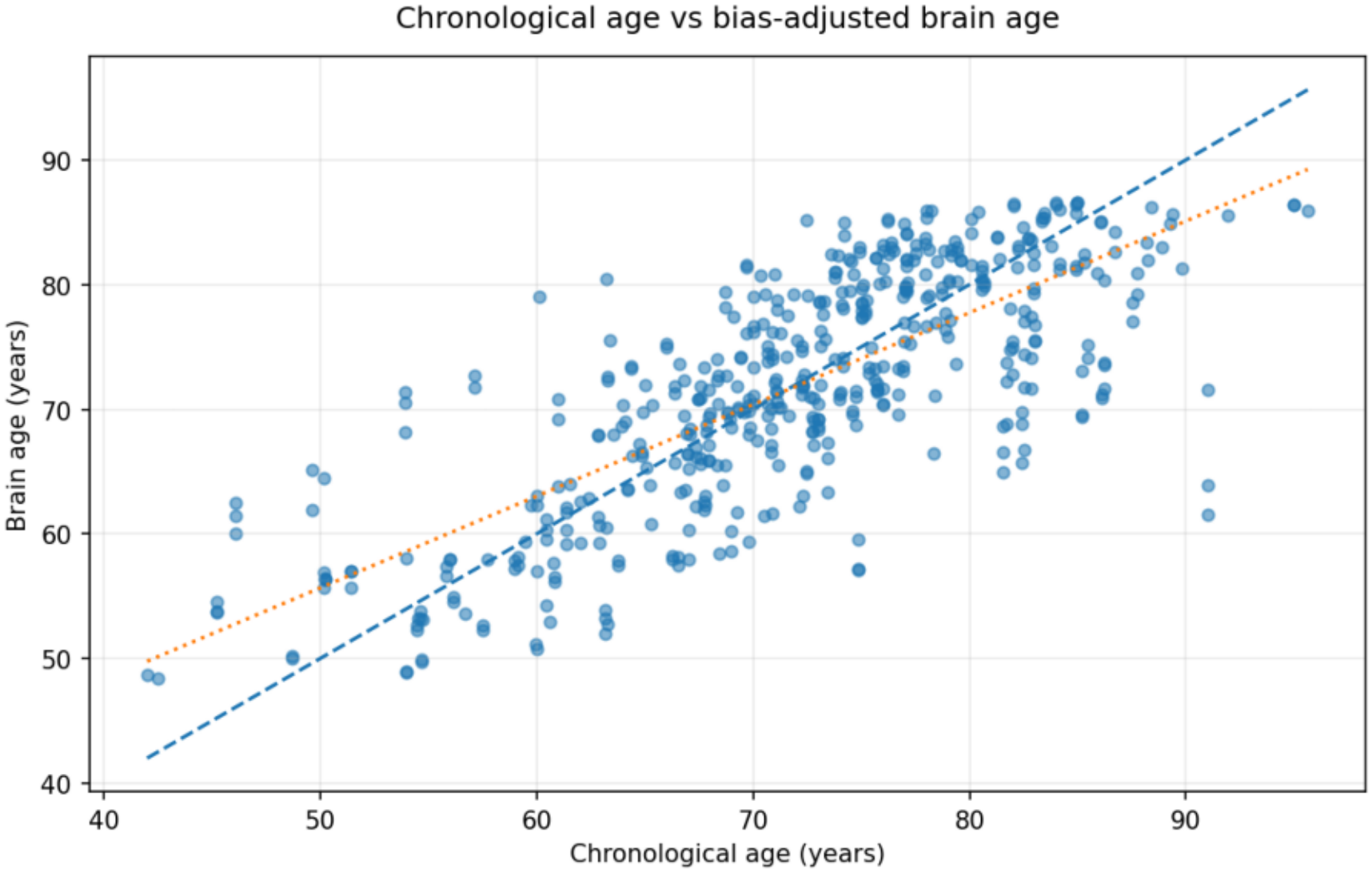

CDR total score distribution
Scan records
0
50
100
150
200
250
300
350
0.0
0.5
Missing
1.0
CDR total score

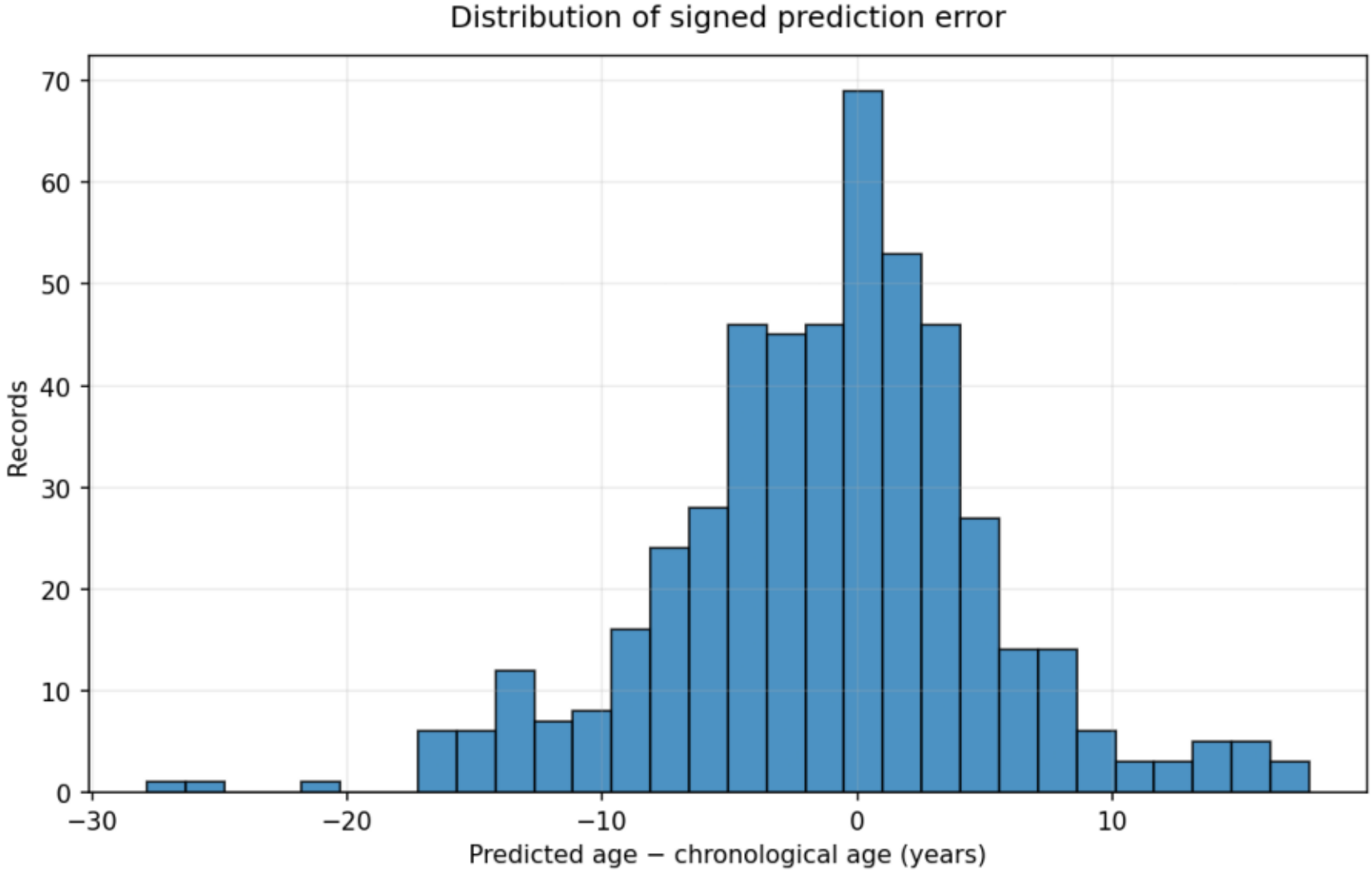

MMSE score distribution
Records
0
20
40
60
80
100
120
140
16
18
20
22
24
26
28
30
MMSE score

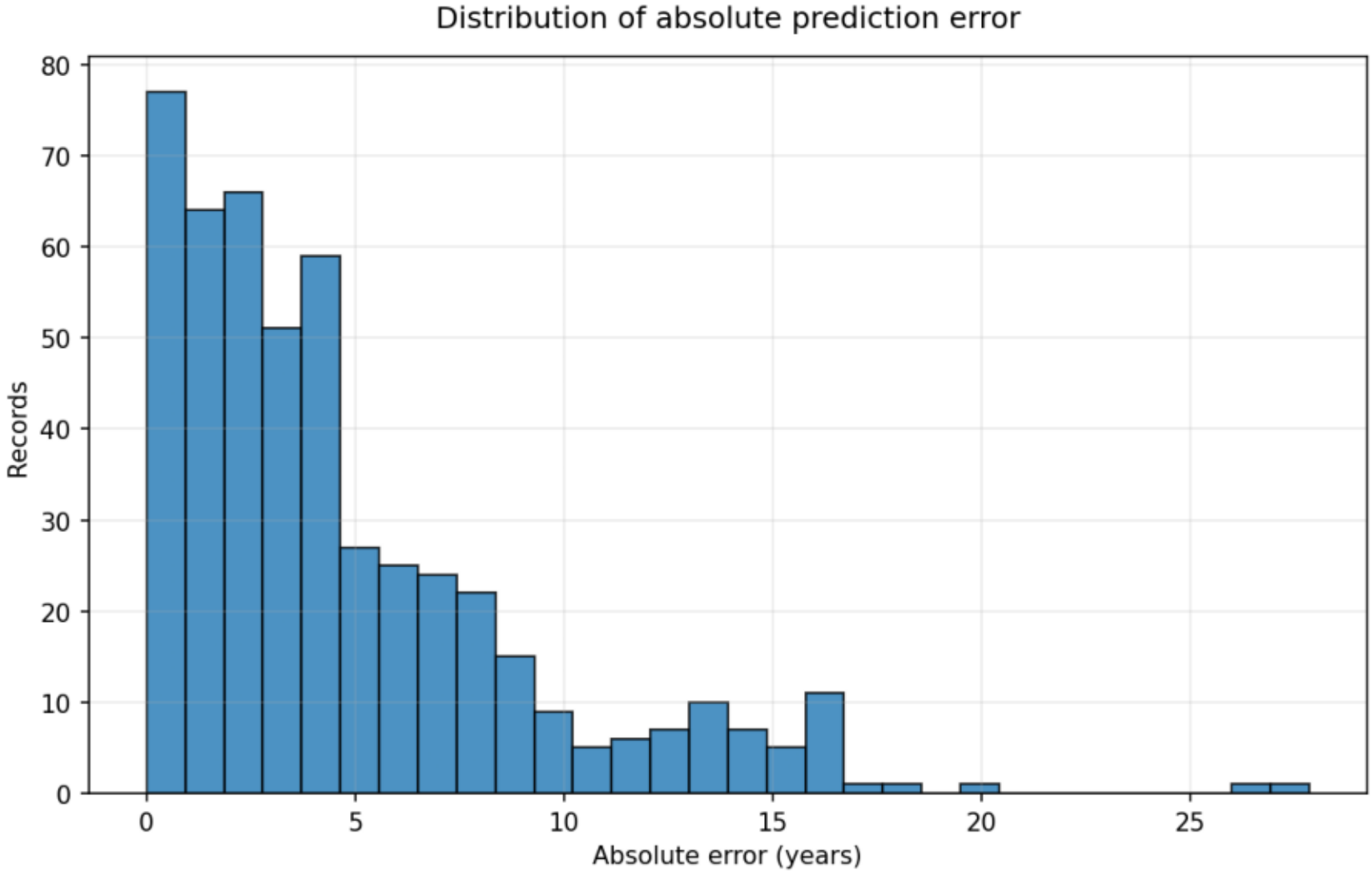

Number of scans per subject
Subjects
0
10
20
30
40
50
1
2
3
4
6
5
7
8
9
10
11
14
Scans per subject

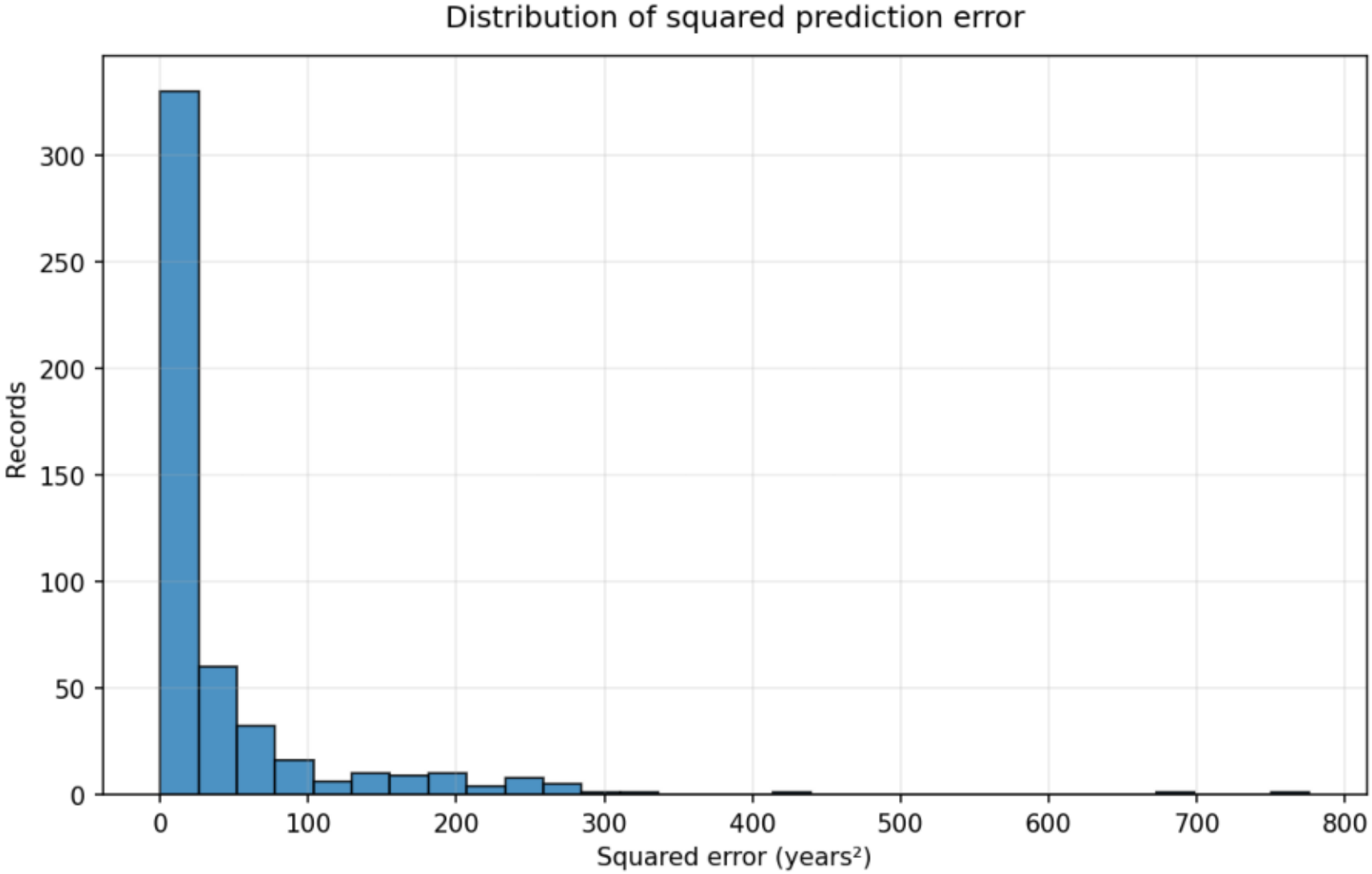

Distribution of scan counts per subject
50
40
30
20
10
0
2
4
6
8
10
12
14

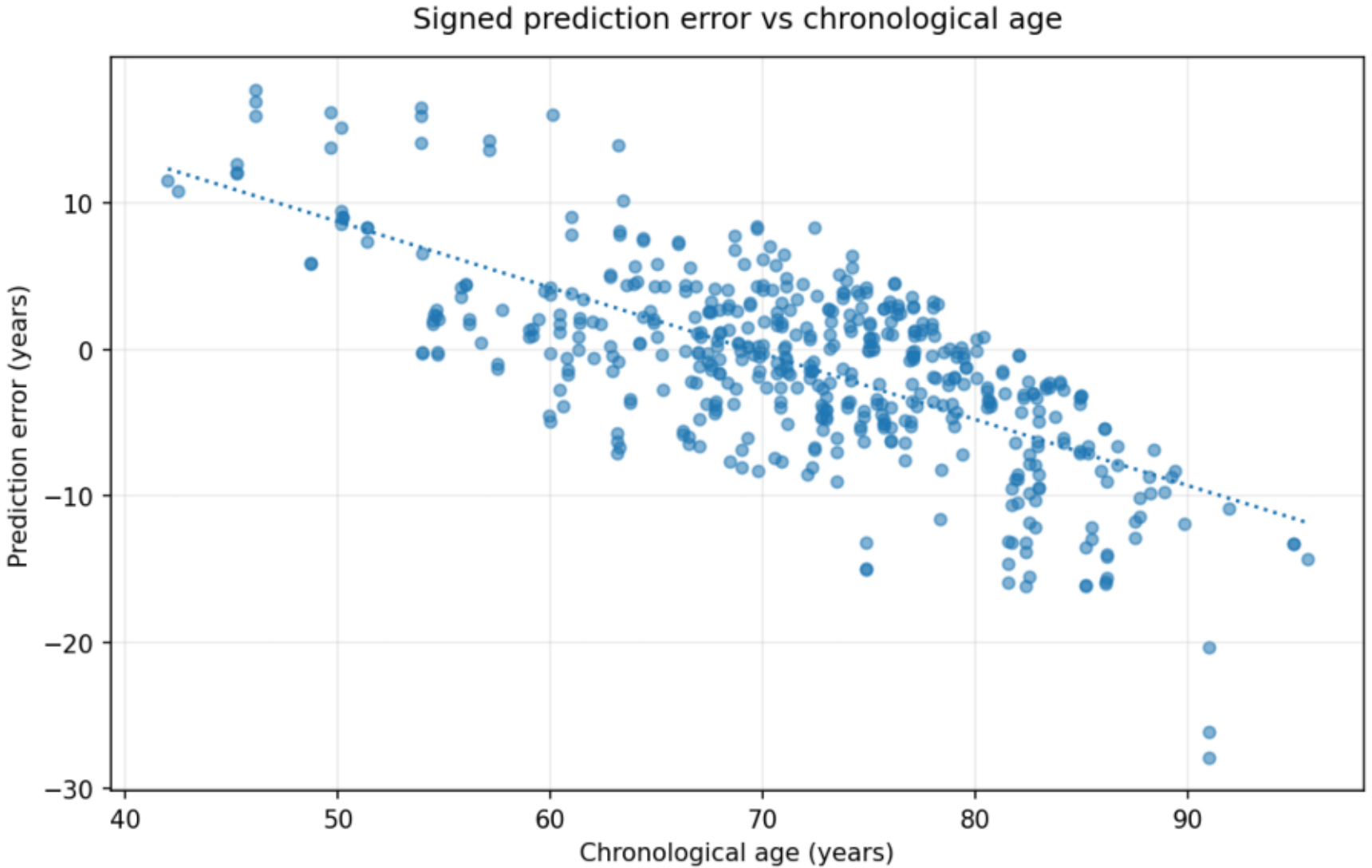

Chronological age vs predicted age
Predicted age (years)
Chronological age (years)
40
50
60
70
80
90

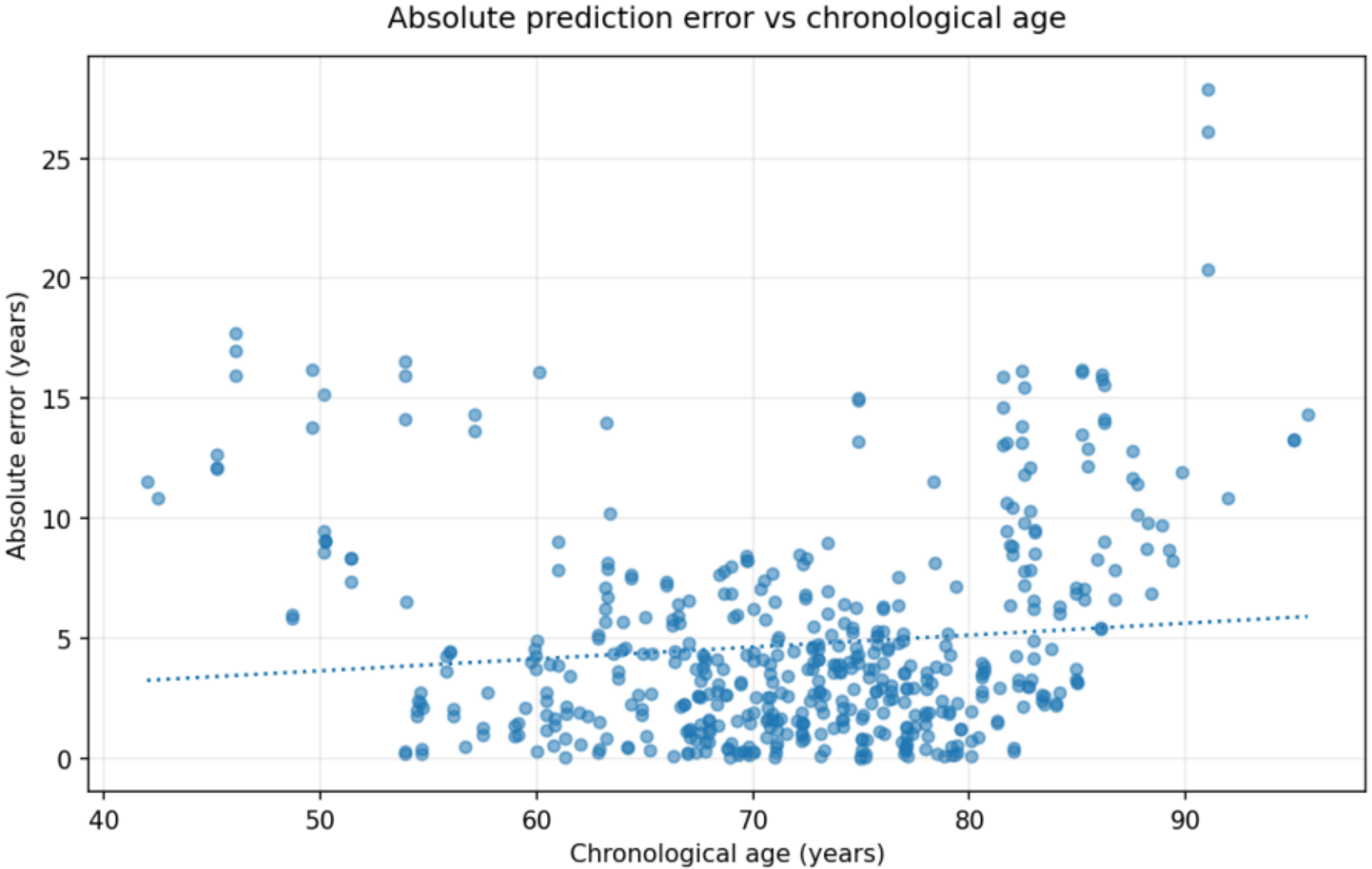

Chronological age vs bias-adjusted brain age
Brain age (years)
Chronological age (years)
40
50
60
70
80
90

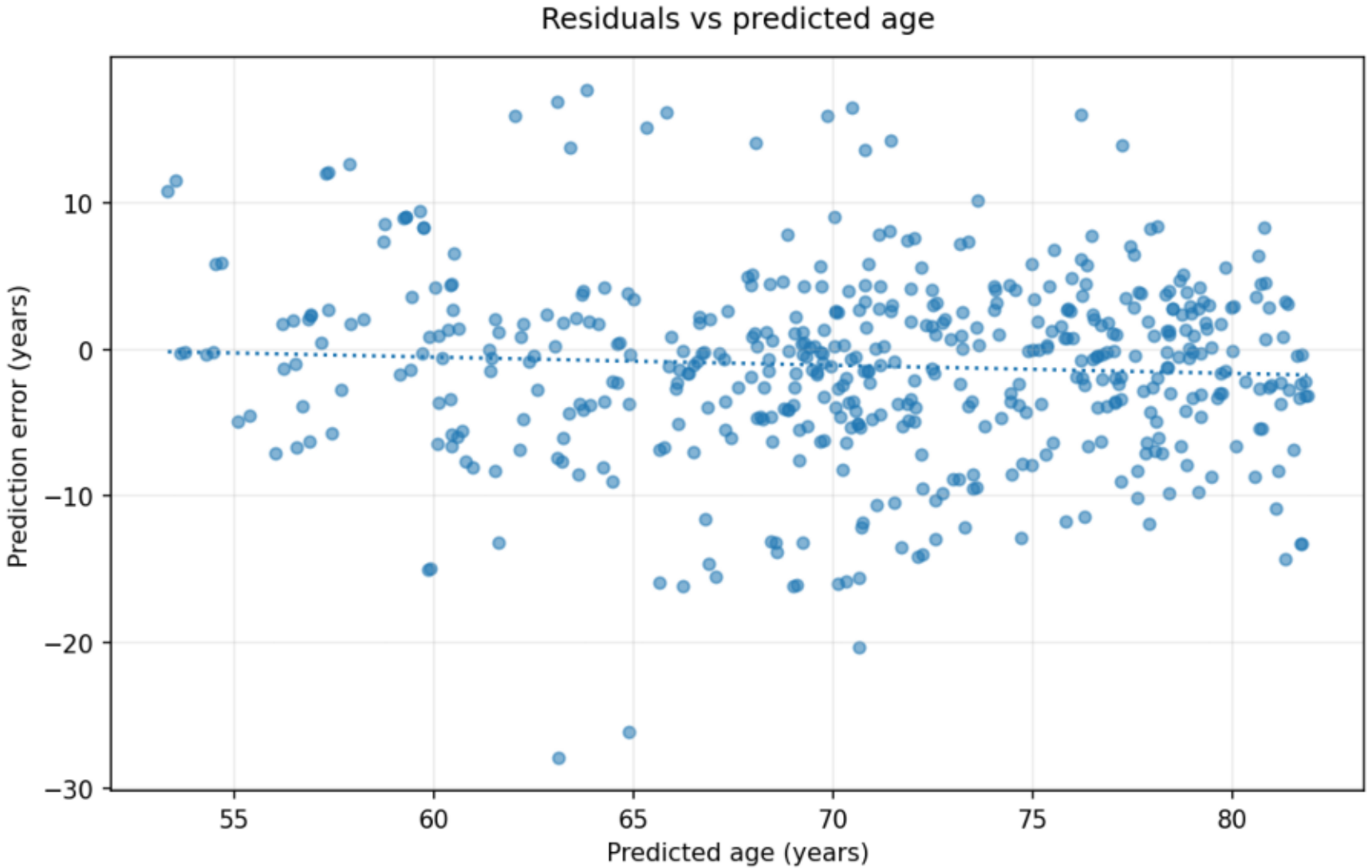

Distribution of signed prediction error
Records
0
10
20
30
40
50
60
70
−30
−20
−10
0
10
Predicted age − chronological age (years)

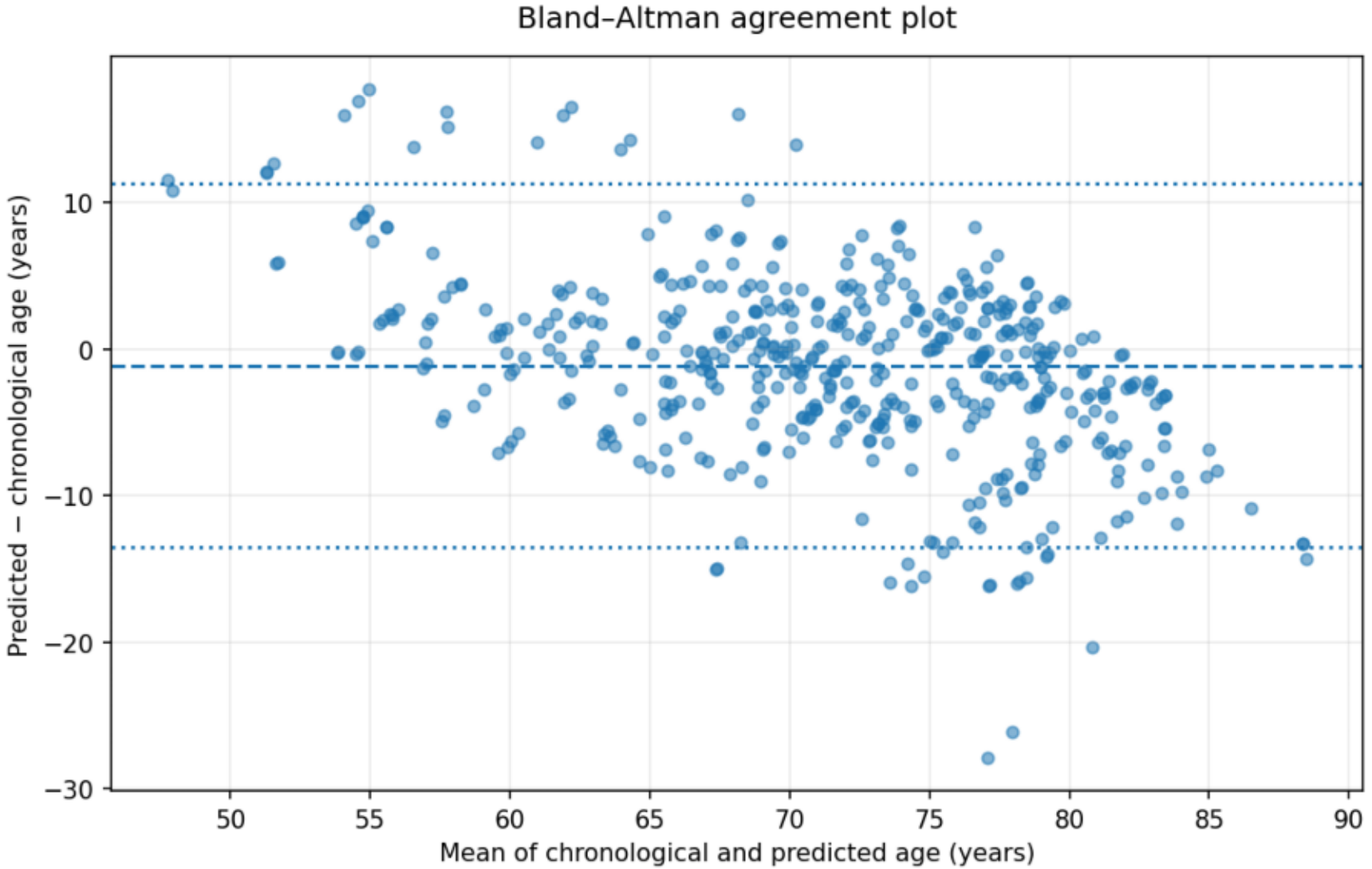

Distribution of absolute prediction error
Records
Absolute error (years)
0
10
20
30
40
50
60
70
80
0
5
10
15
20
25

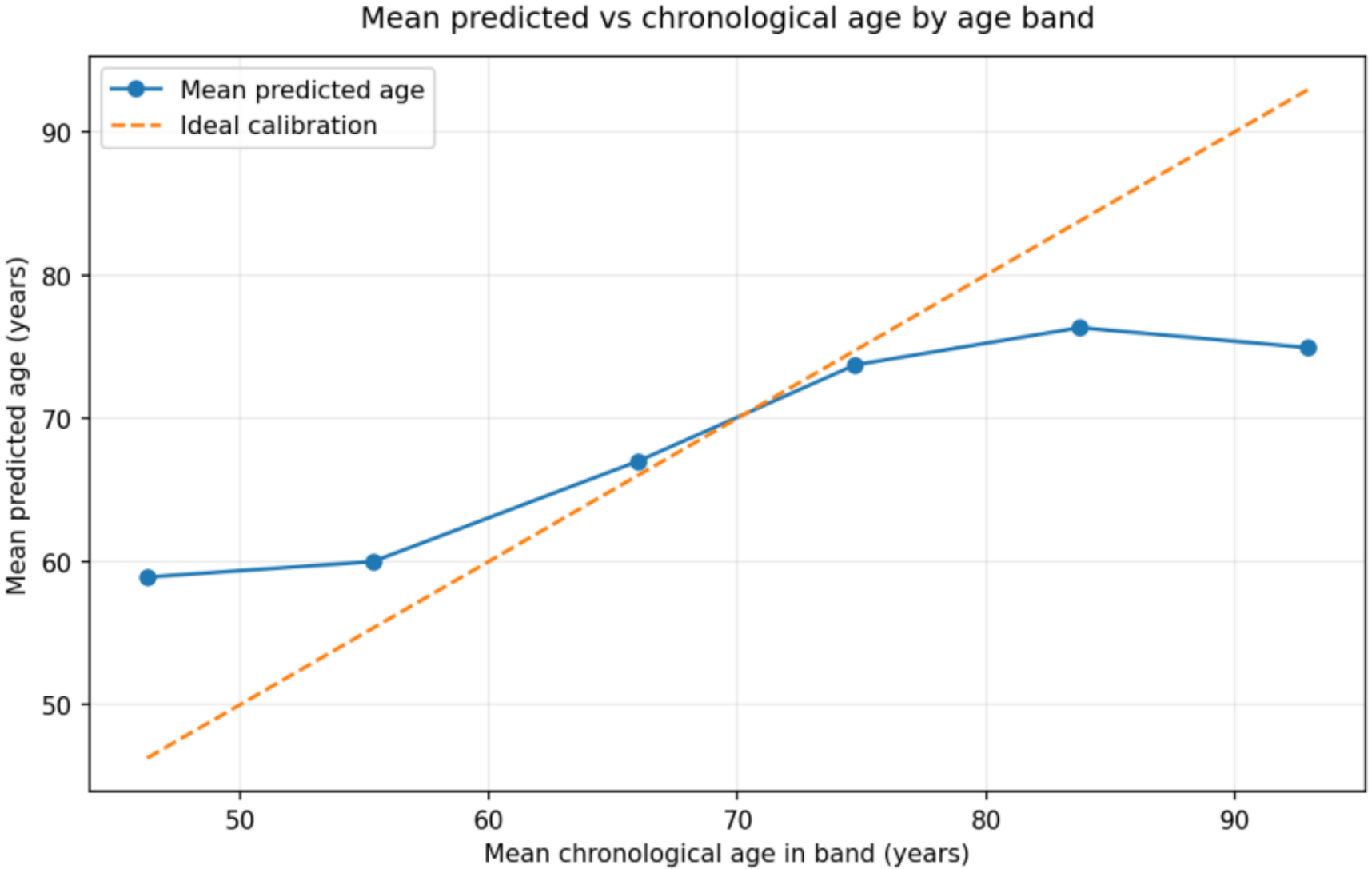

Distribution of squared prediction error
Records
0
50
100
150
200
250
300
0
100
200
300
400
500
600
700
800
Squared error (years²)

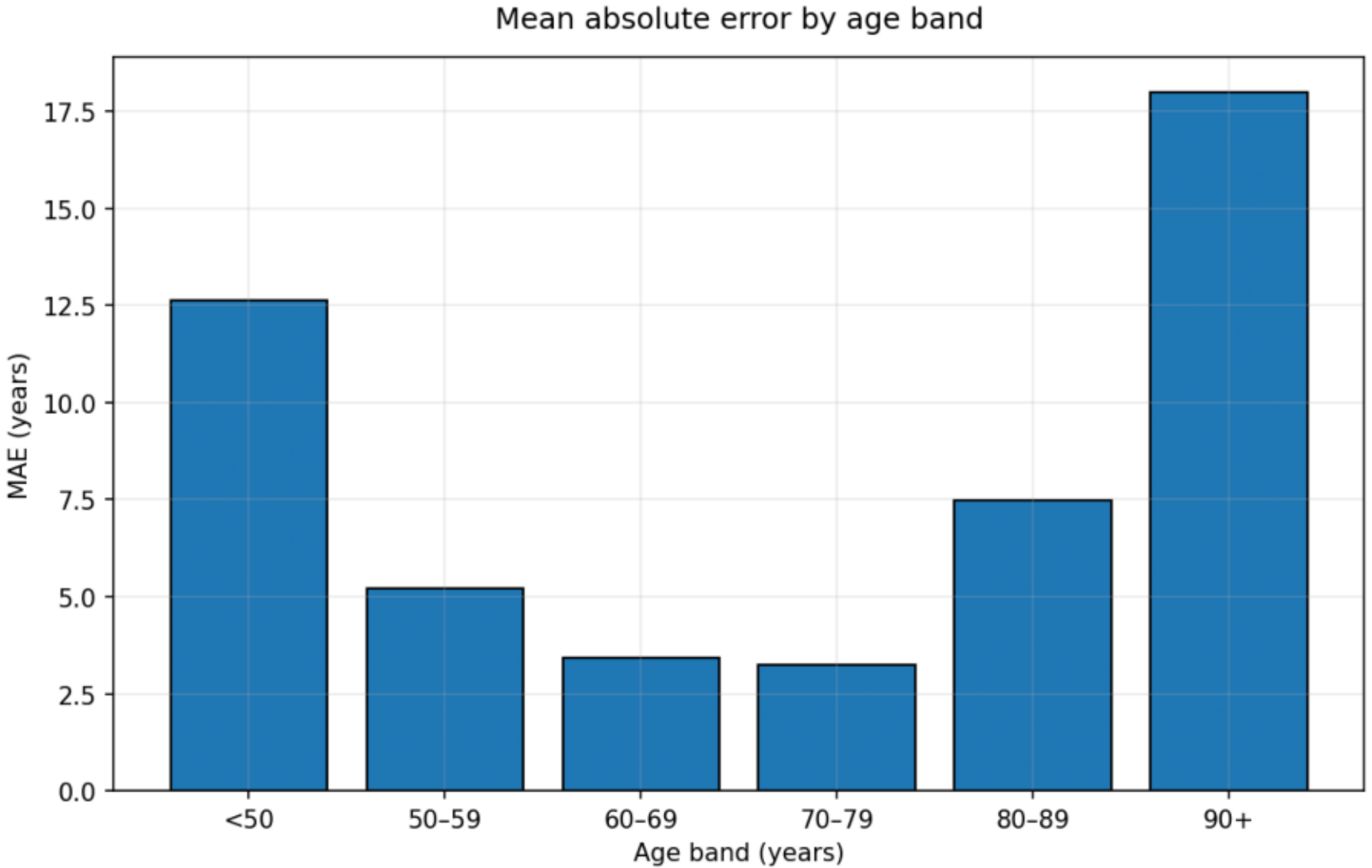

Signed prediction error vs chronological age
Prediction error (years)
10
0
−10
−20
−30
40
50
60
70
80
90
Chronological age (years)

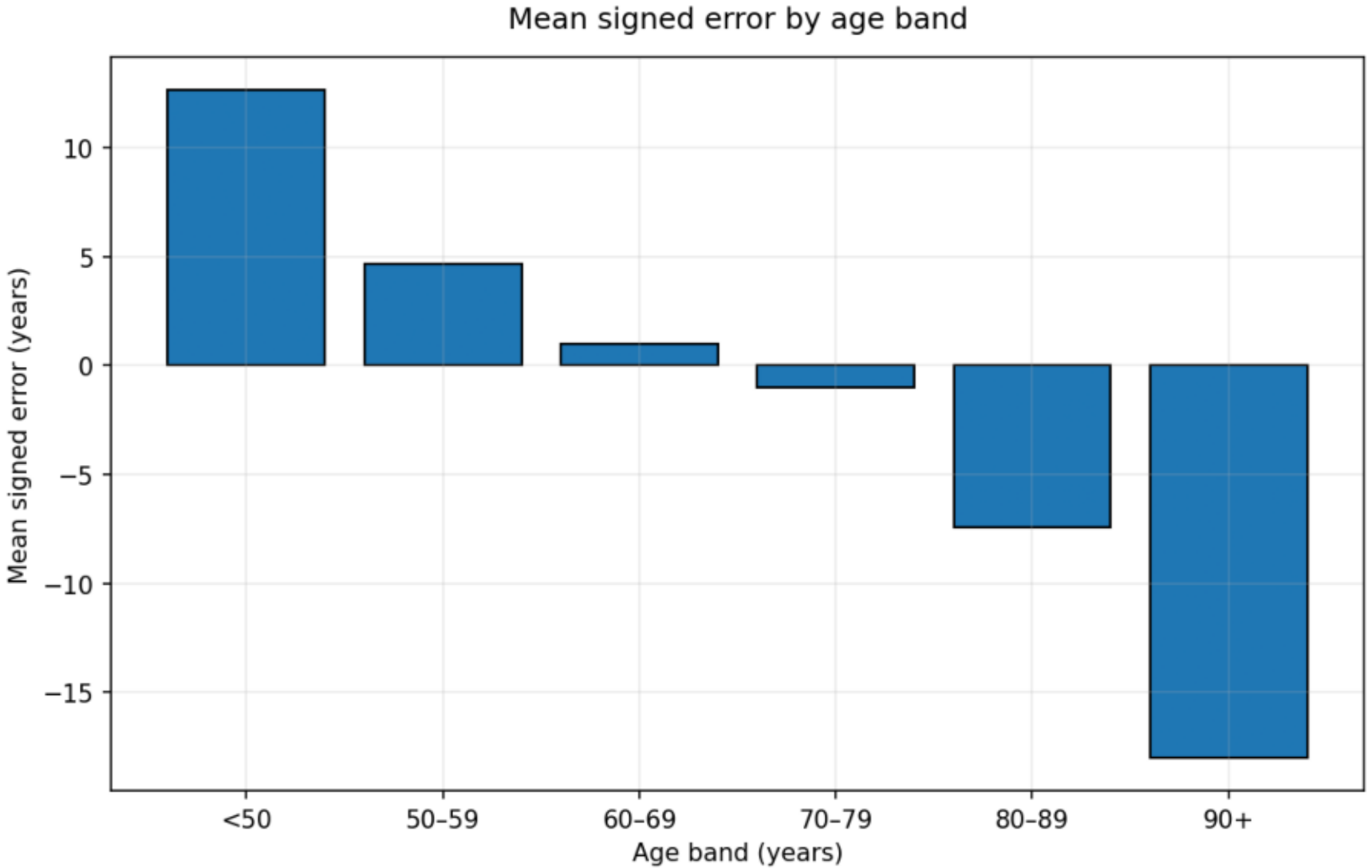

Absolute prediction error vs chronological age
Absolute error (years)
0
5
10
15
20
25
40
50
60
70
80
90
Chronological age (years)

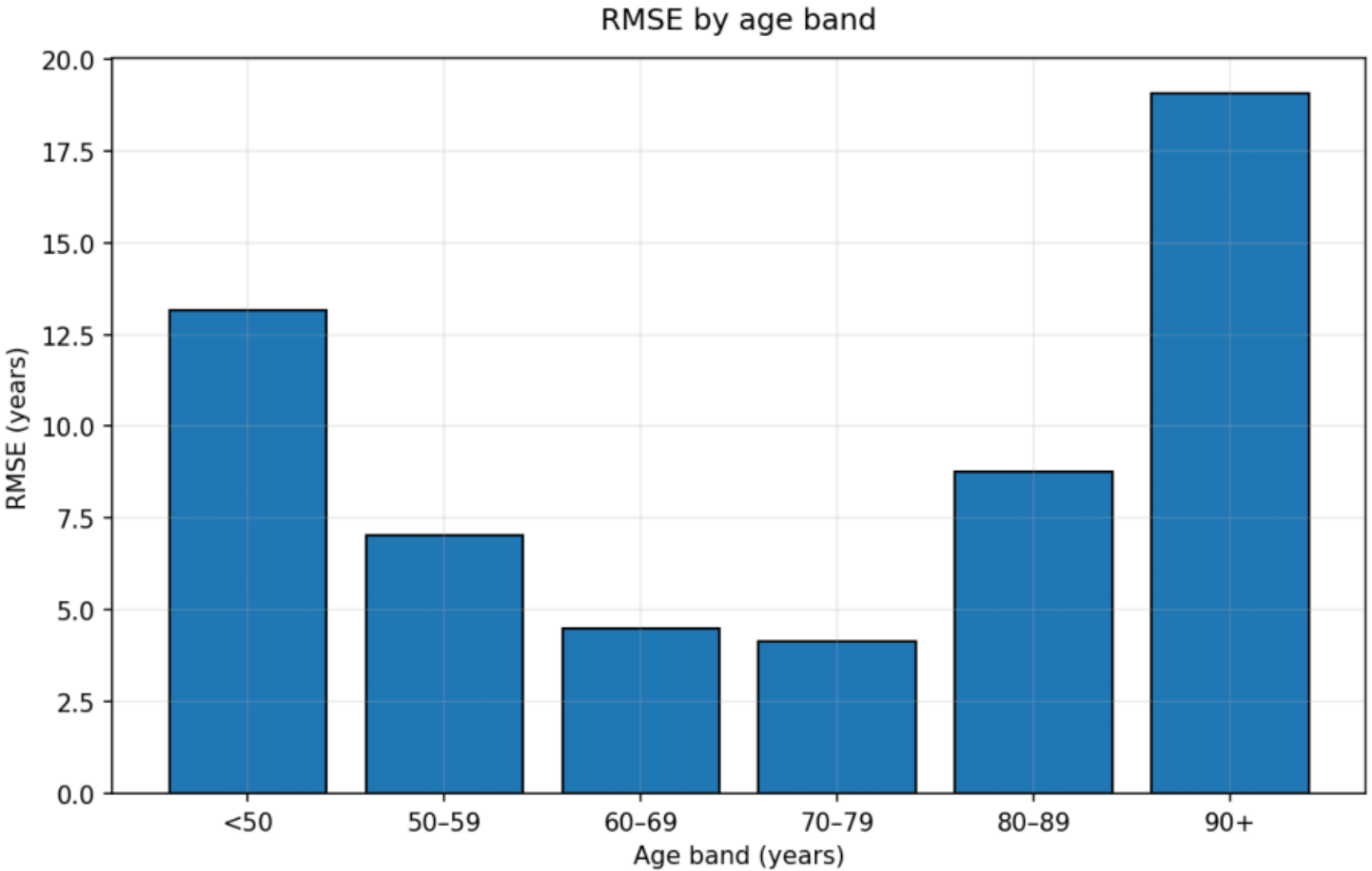

Residuals vs predicted age
Prediction error (years)
10
0
−10
−20
−30
55
60
65
70
75
80
Predicted age (years)

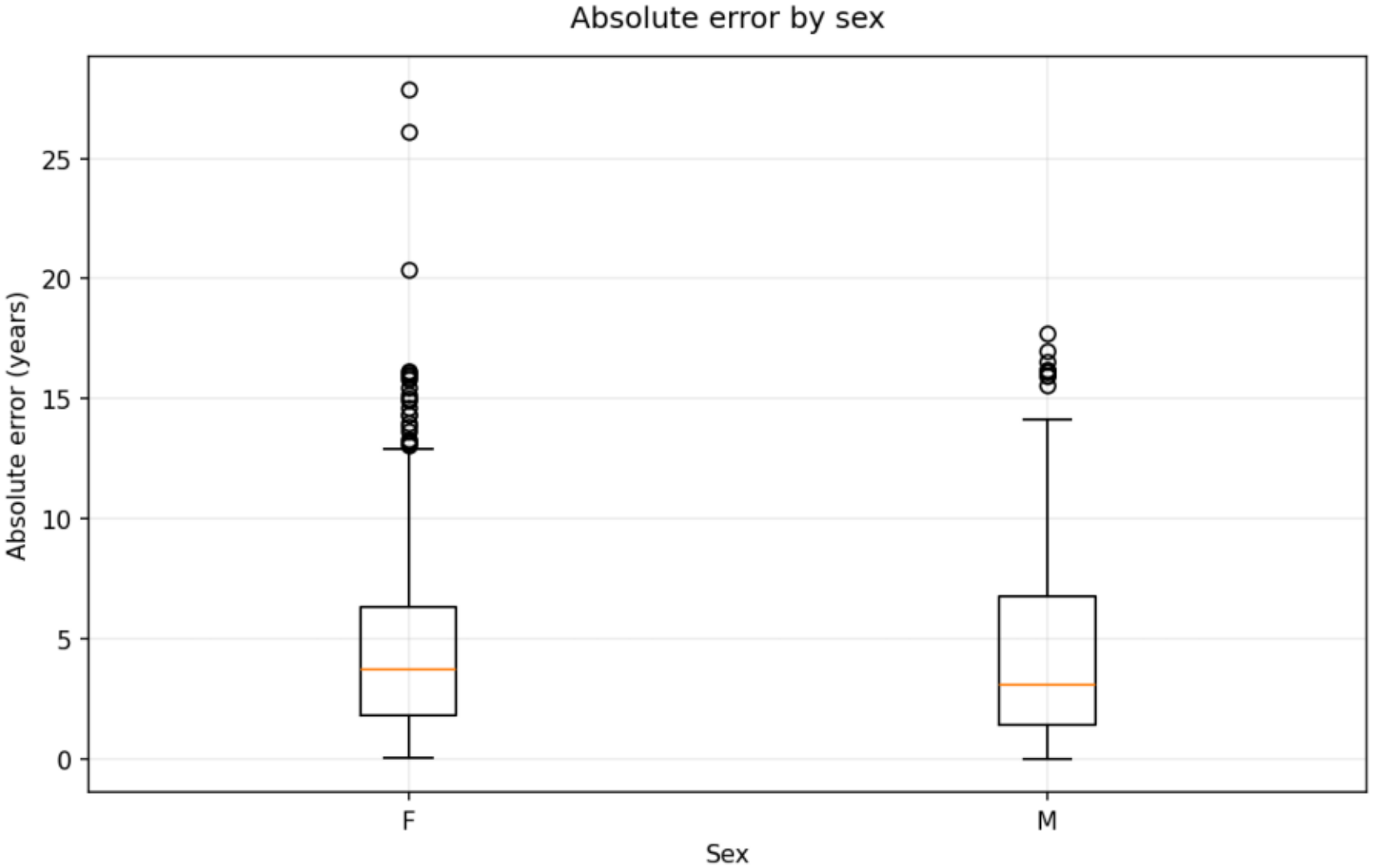

Bland–Altman agreement plot
Predicted − chronological age (years)
10
0
−10
−20
−30
50
55
60
65
70
75
80
85
90
Mean of chronological and predicted age (years)

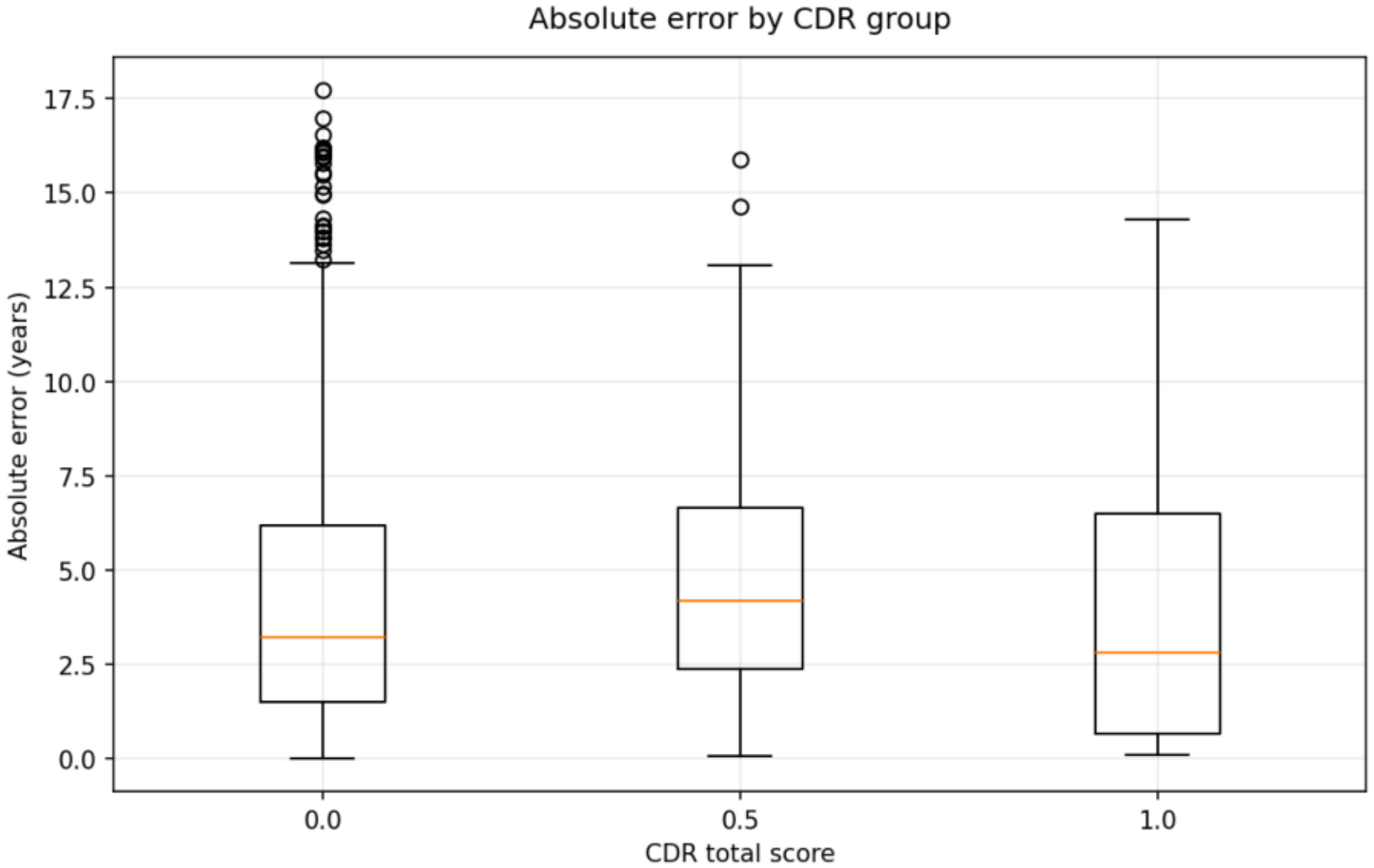

Mean predicted vs chronological age by age band
Mean predicted age
Ideal calibration
Mean predicted age (years)
90
80
70
60
50
50
60
70
80
90
Mean chronological age in band (years)

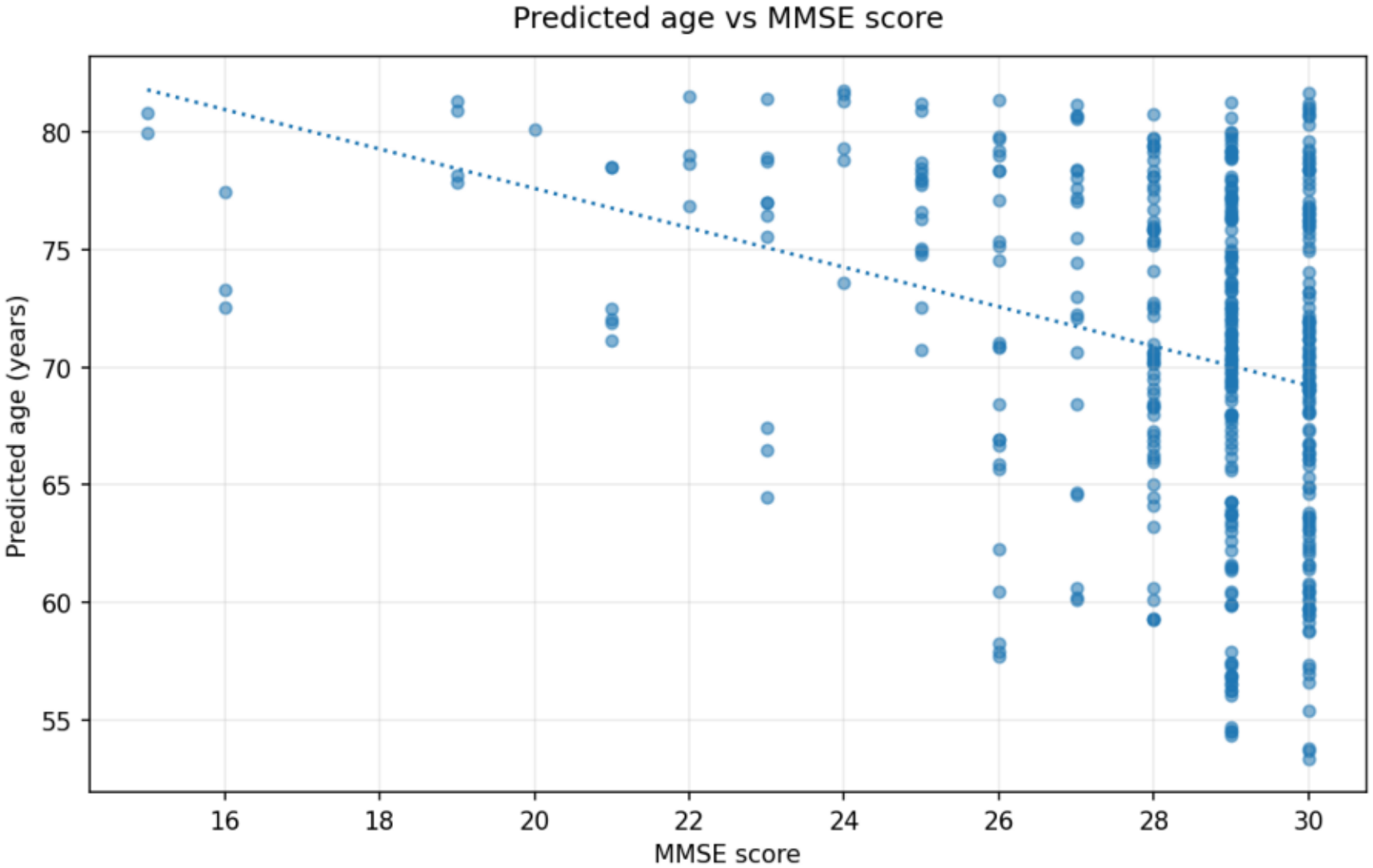

Mean absolute error by age band
MAE (years)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
<50
50–59
60–69
70–79
80–89
90+
Age band (years)

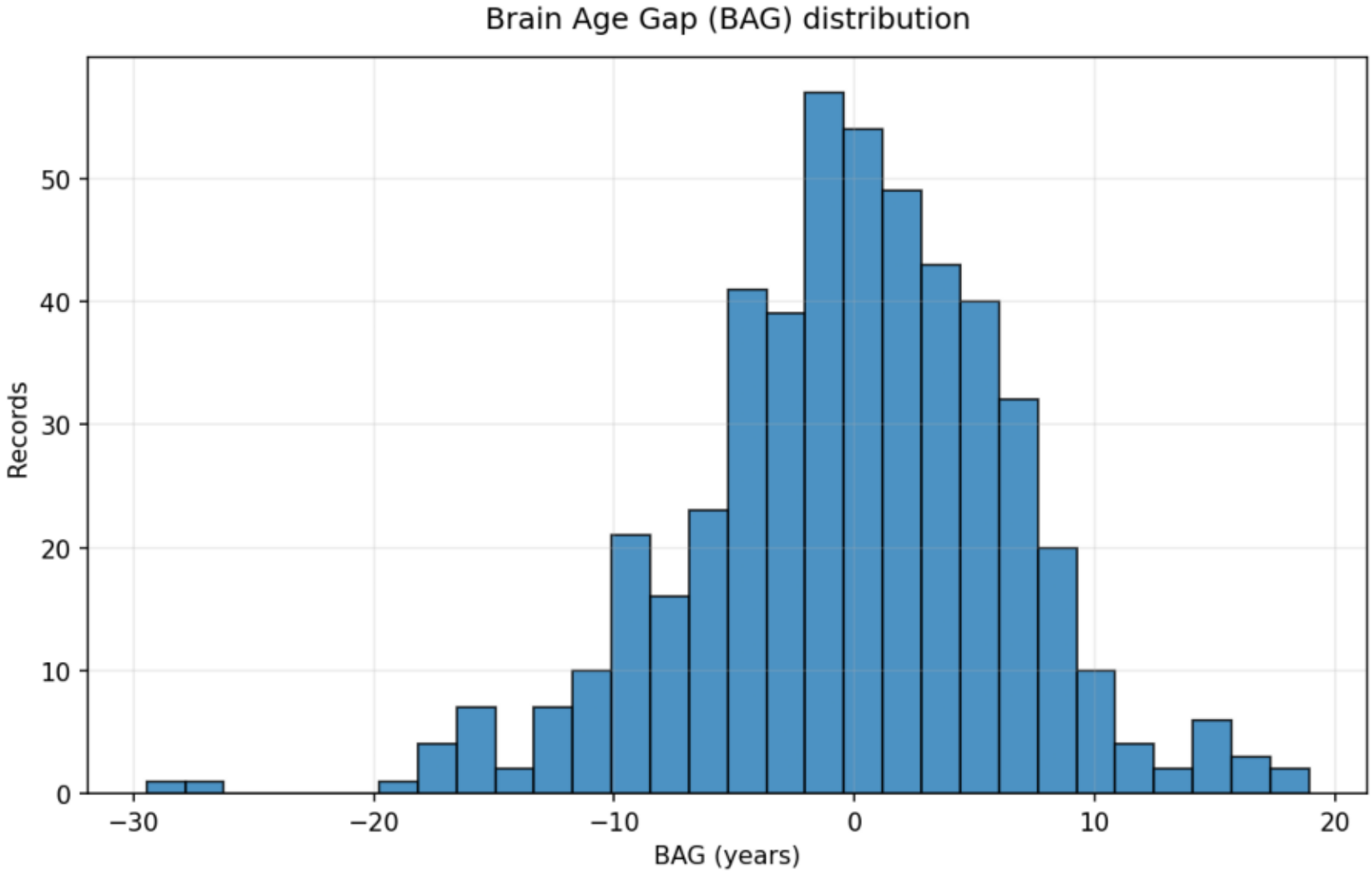

Mean signed error by age band
Mean signed error (years)
10
5
0
−5
−10
−15
<50
50–59
60–69
70–79
80–89
90+
Age band (years)

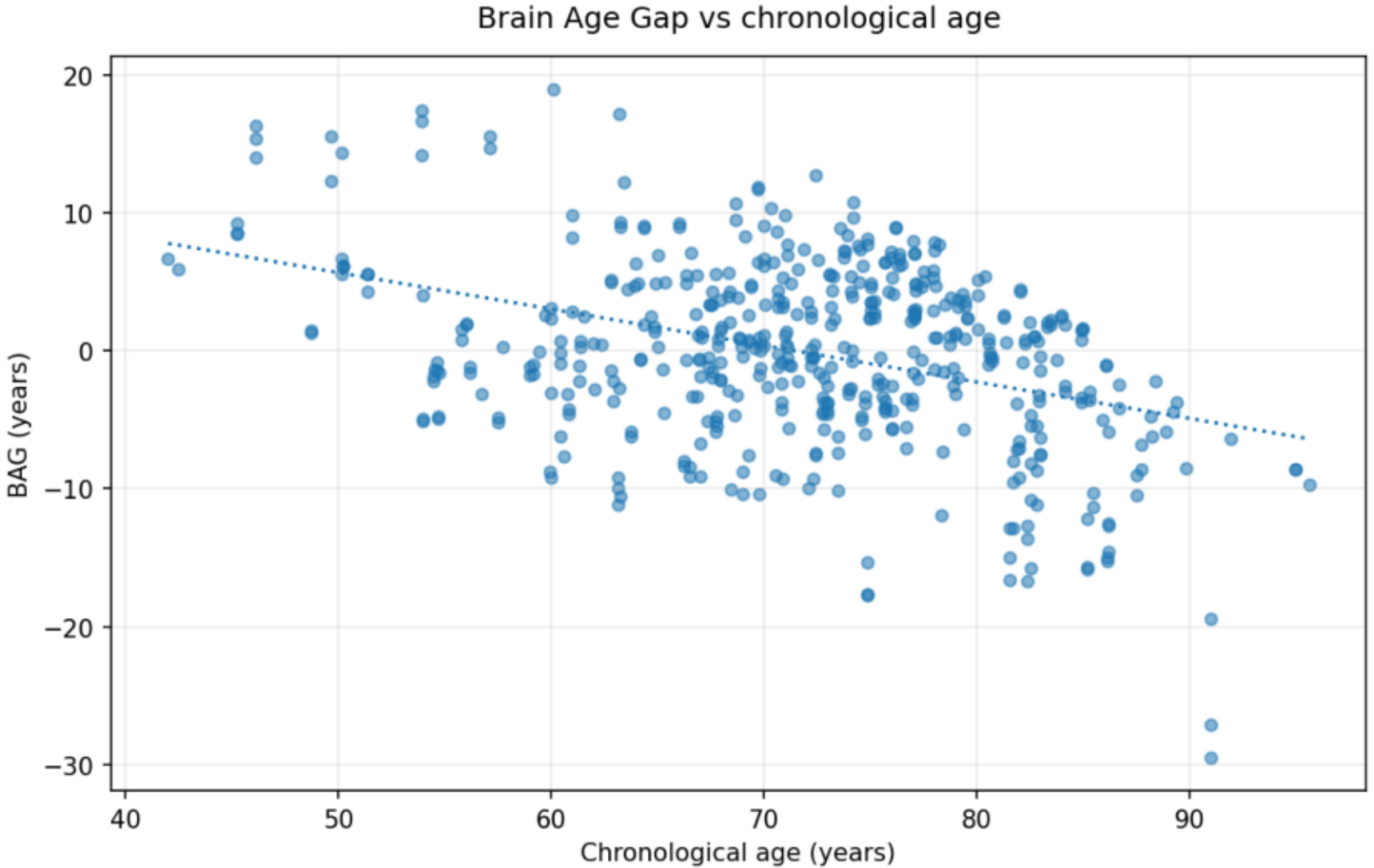

RMSE by age band
RMSE (years)
20.0
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
<50
50–59
60–69
70–79
80–89
90+
Age band (years)

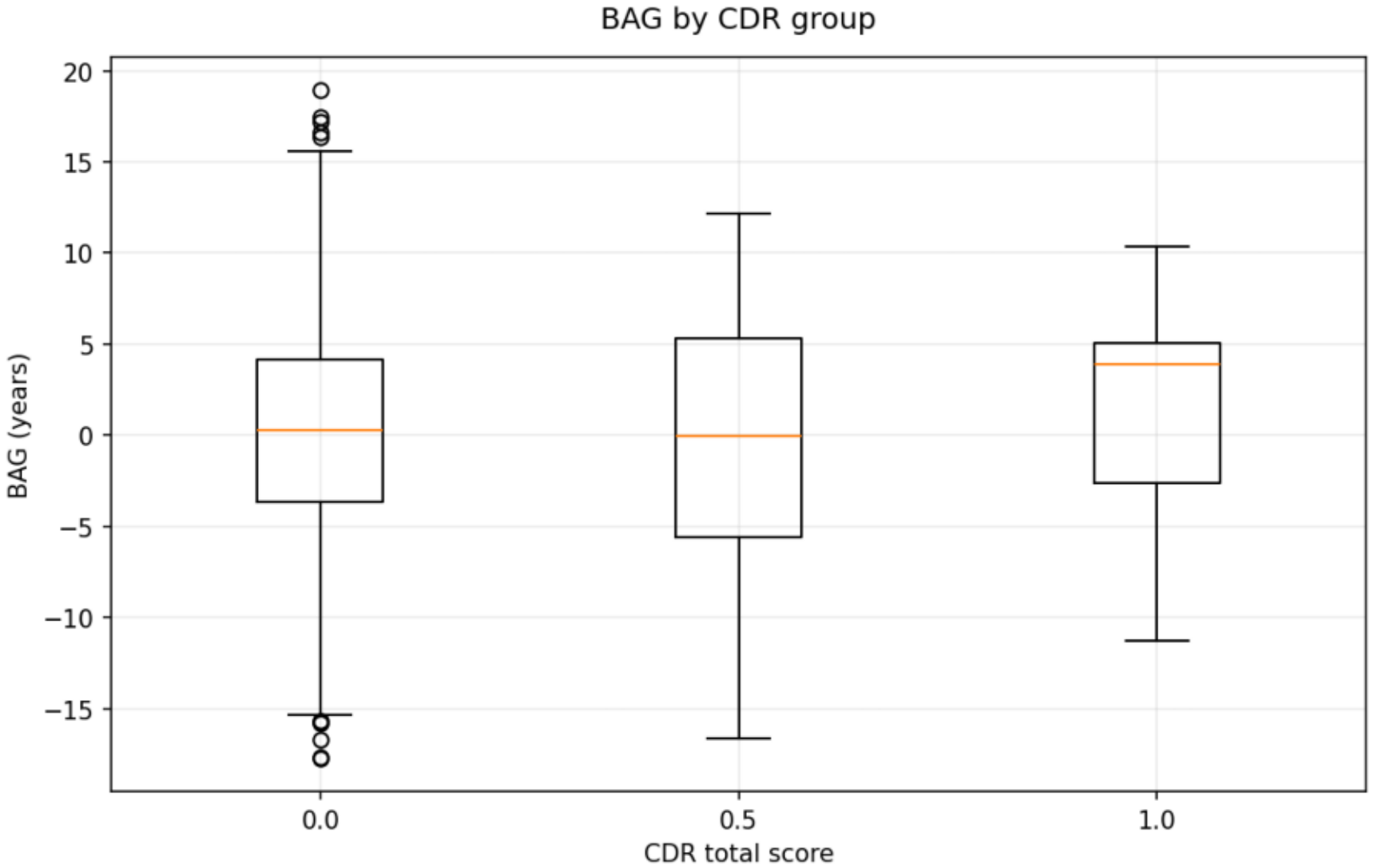

Absolute error by sex
Absolute error (years)
0
5
10
15
20
25
F
M
Sex

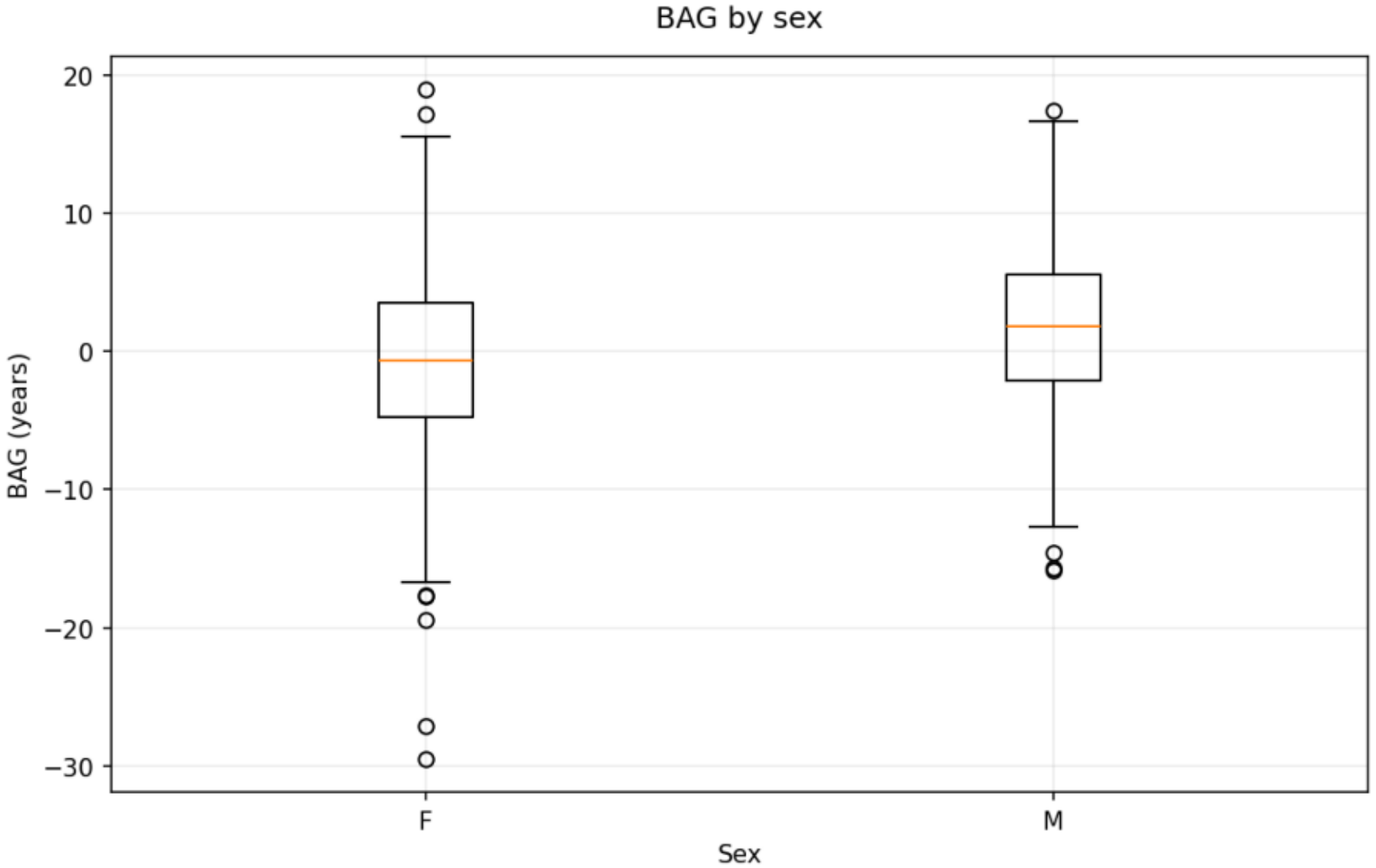

Absolute error by CDR group
Absolute error (years)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0.0
0.5
1.0
CDR total score

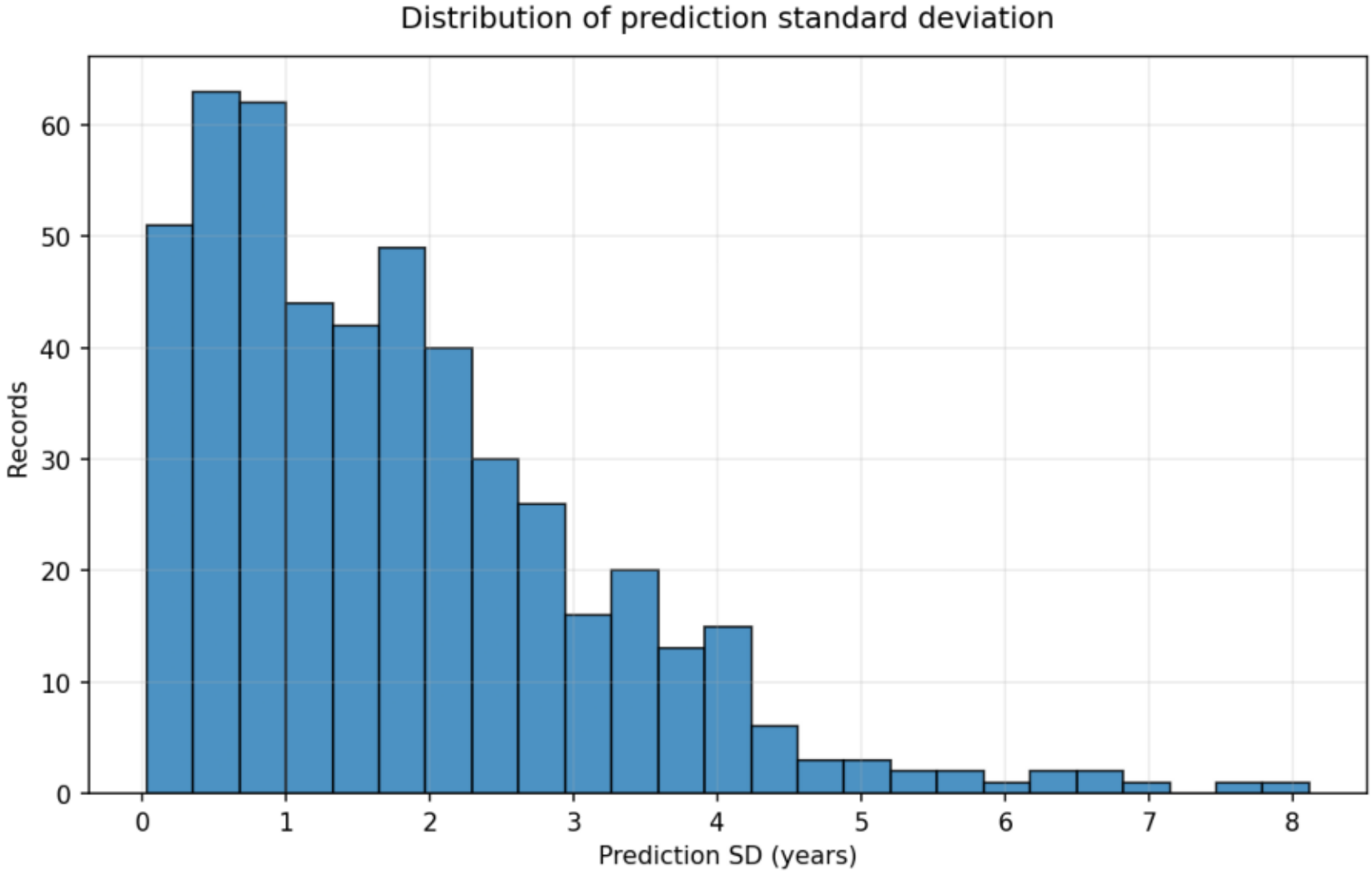

Predicted age vs MMSE score
Predicted age (years)
80
75
70
65
60
55
16
18
20
22
24
26
28
30
MMSE score

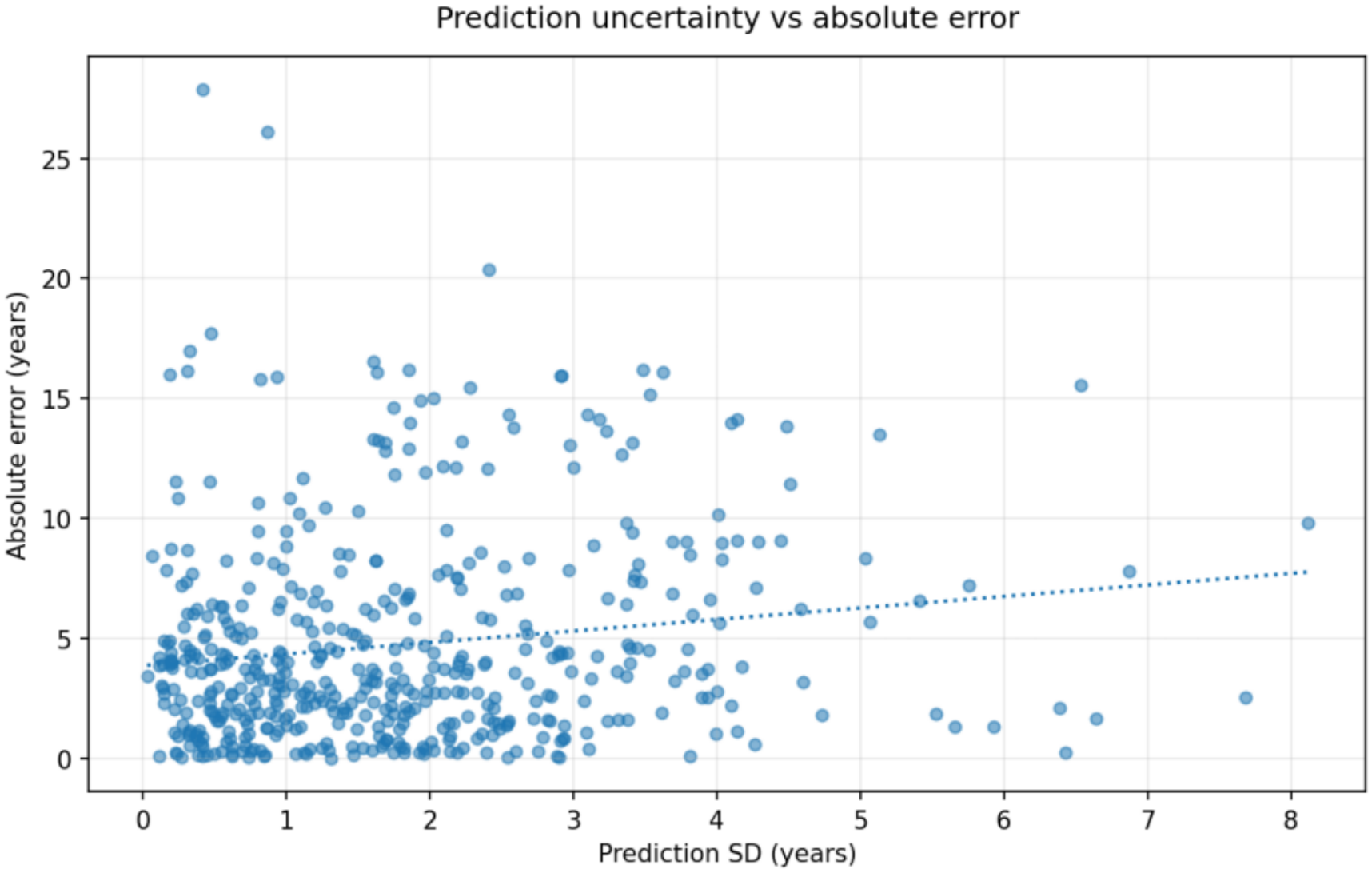

Brain Age Gap (BAG) distribution
Records
0
10
20
30
40
50
−30
−20
−10
0
10
20
BAG (years)

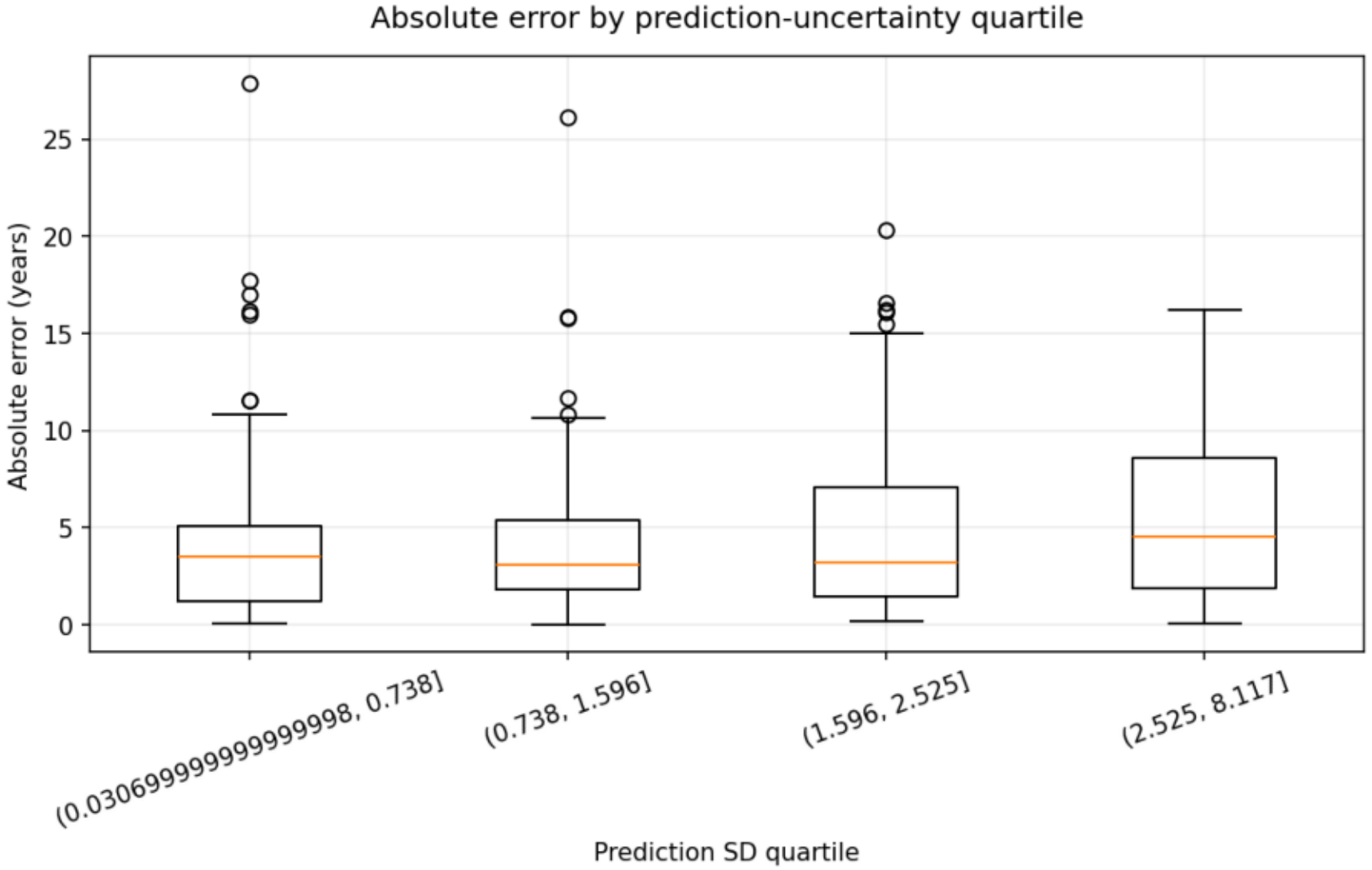

Brain Age Gap vs chronological age
BAG (years)
20
10
0
−10
−20
−30
40
50
60
70
80
90
Chronological age (years)

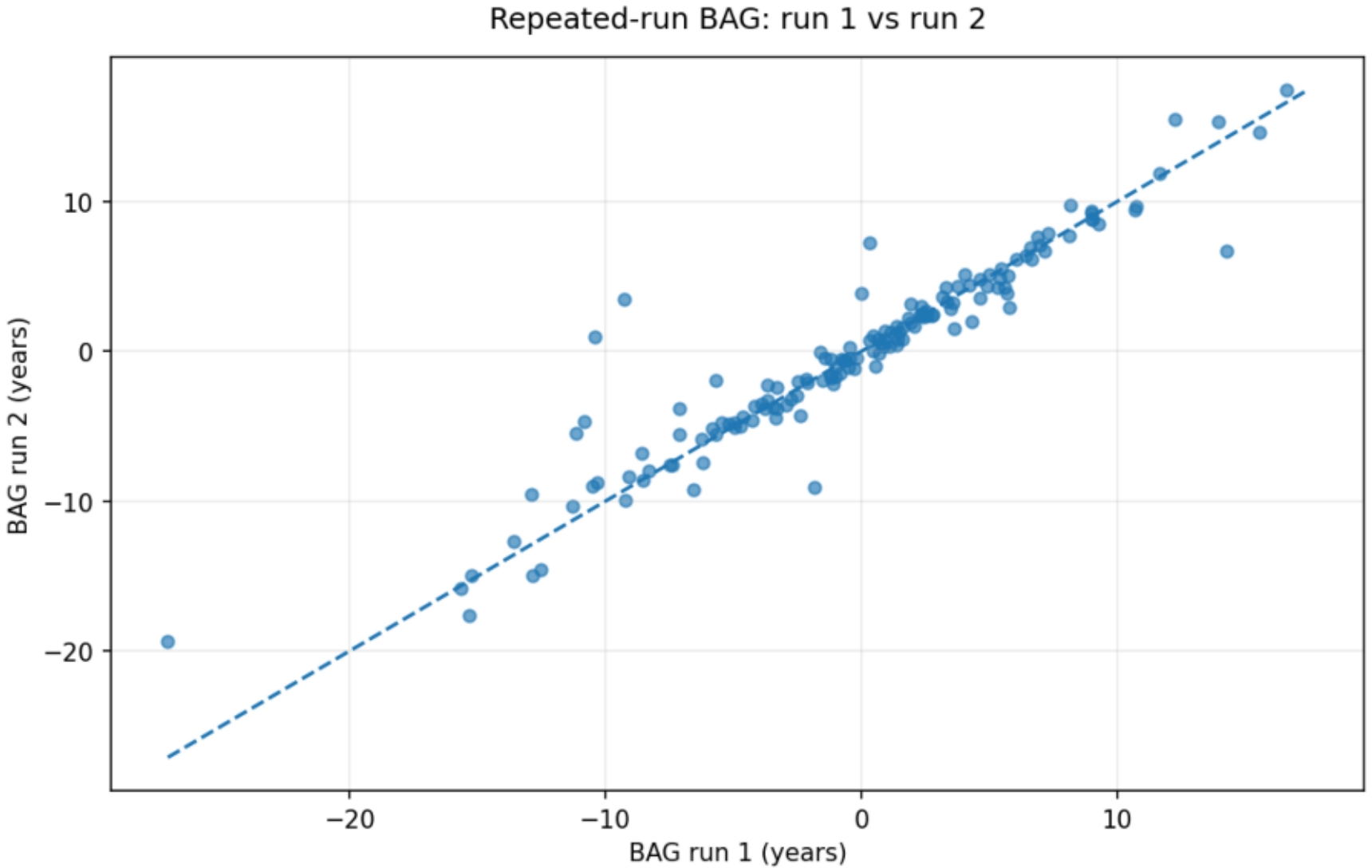

BAG by CDR group
BAG (years)
20
15
10
5
0
−5
−10
−15
0.0
0.5
1.0
CDR total score

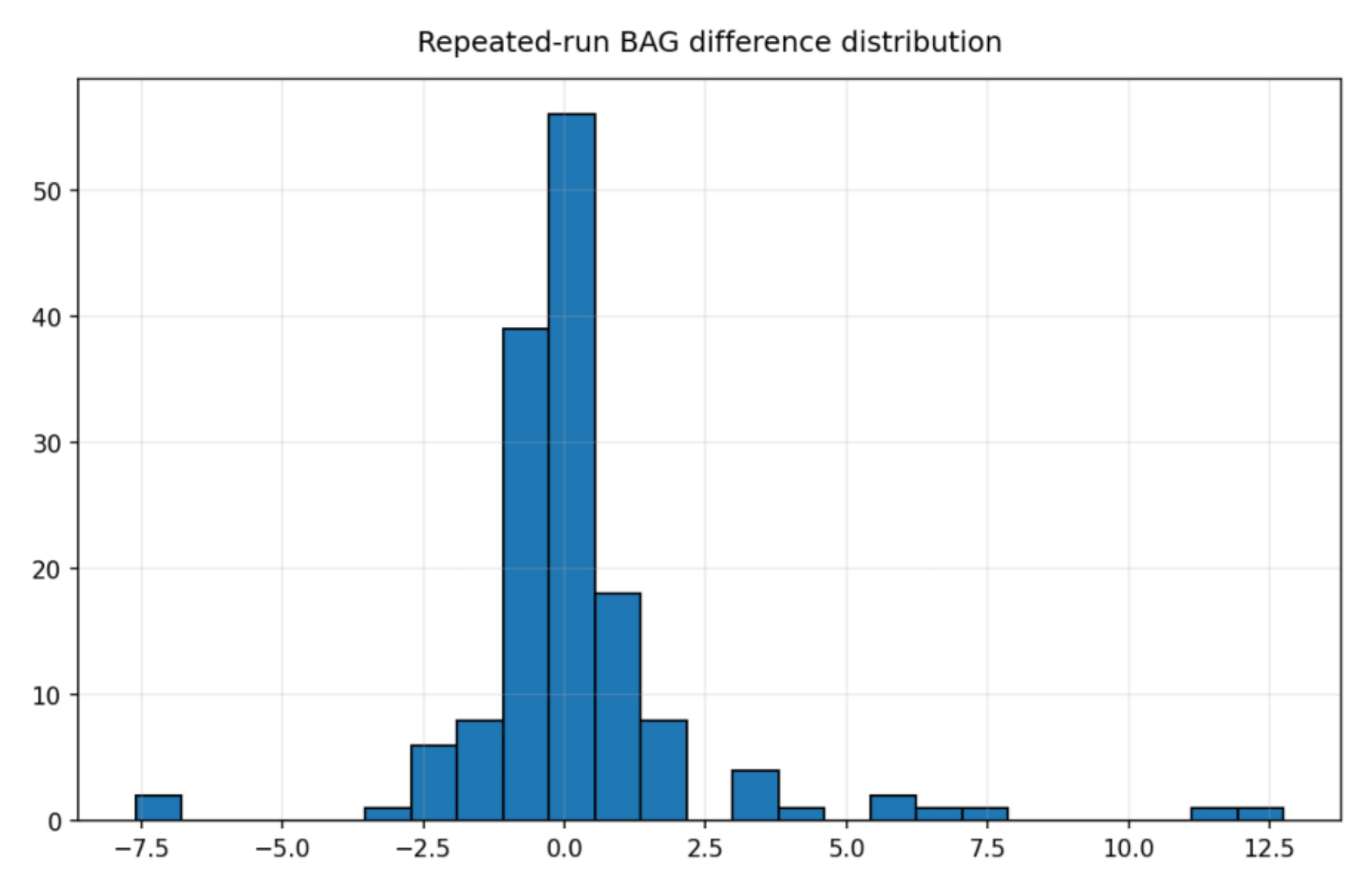

BAG by sex
BAG (years)
20
10
0
−10
−20
−30
F
M
Sex

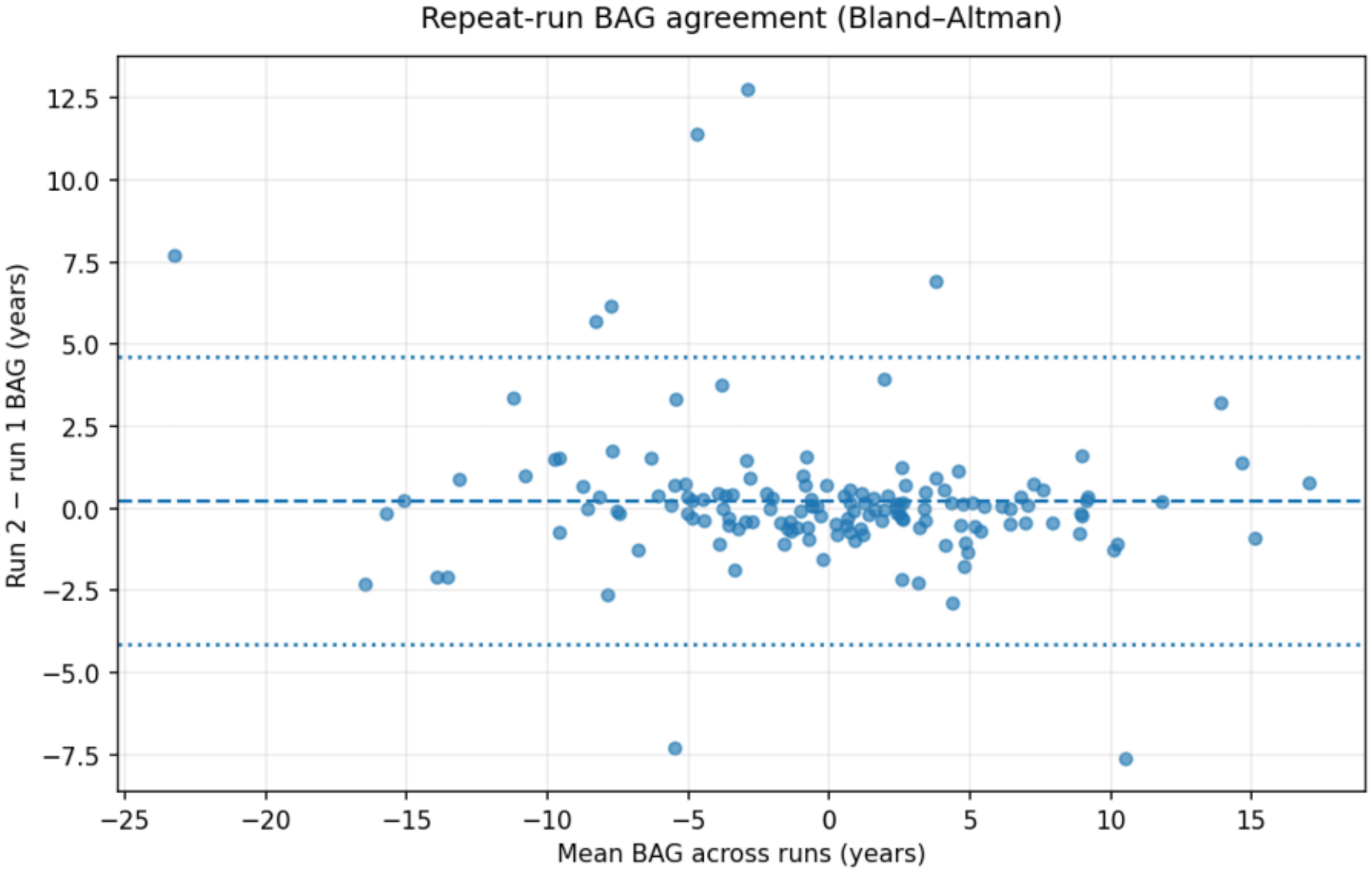

Distribution of prediction standard deviation
Records
Prediction SD (years)
0
10
20
30
40
50
60
0
1
2
3
4
5
6
7
8

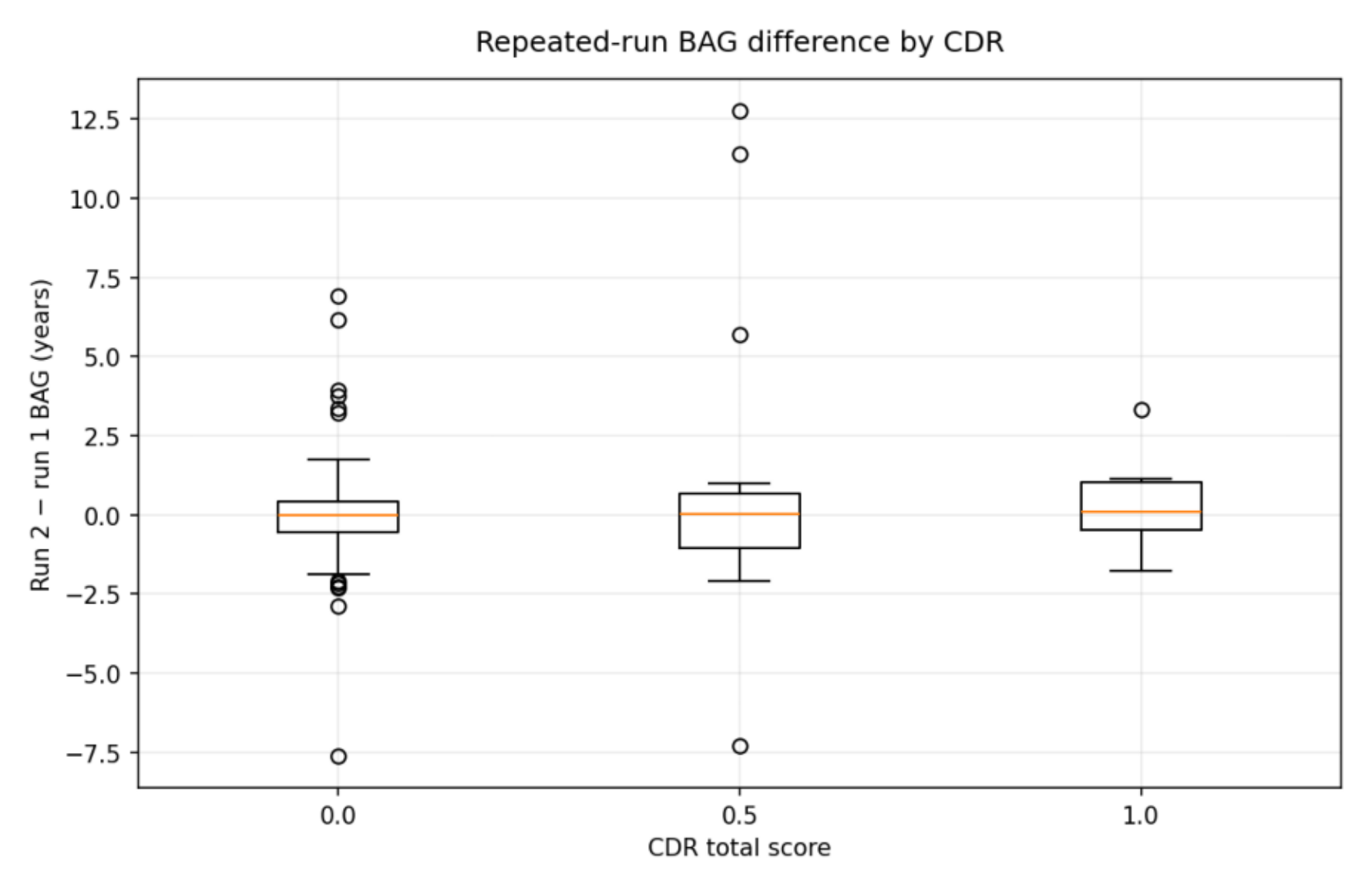

Prediction uncertainty vs absolute error
Absolute error (years)
Prediction SD (years)
0
5
10
15
20
25
0
1
2
3
4
5
6
7
8

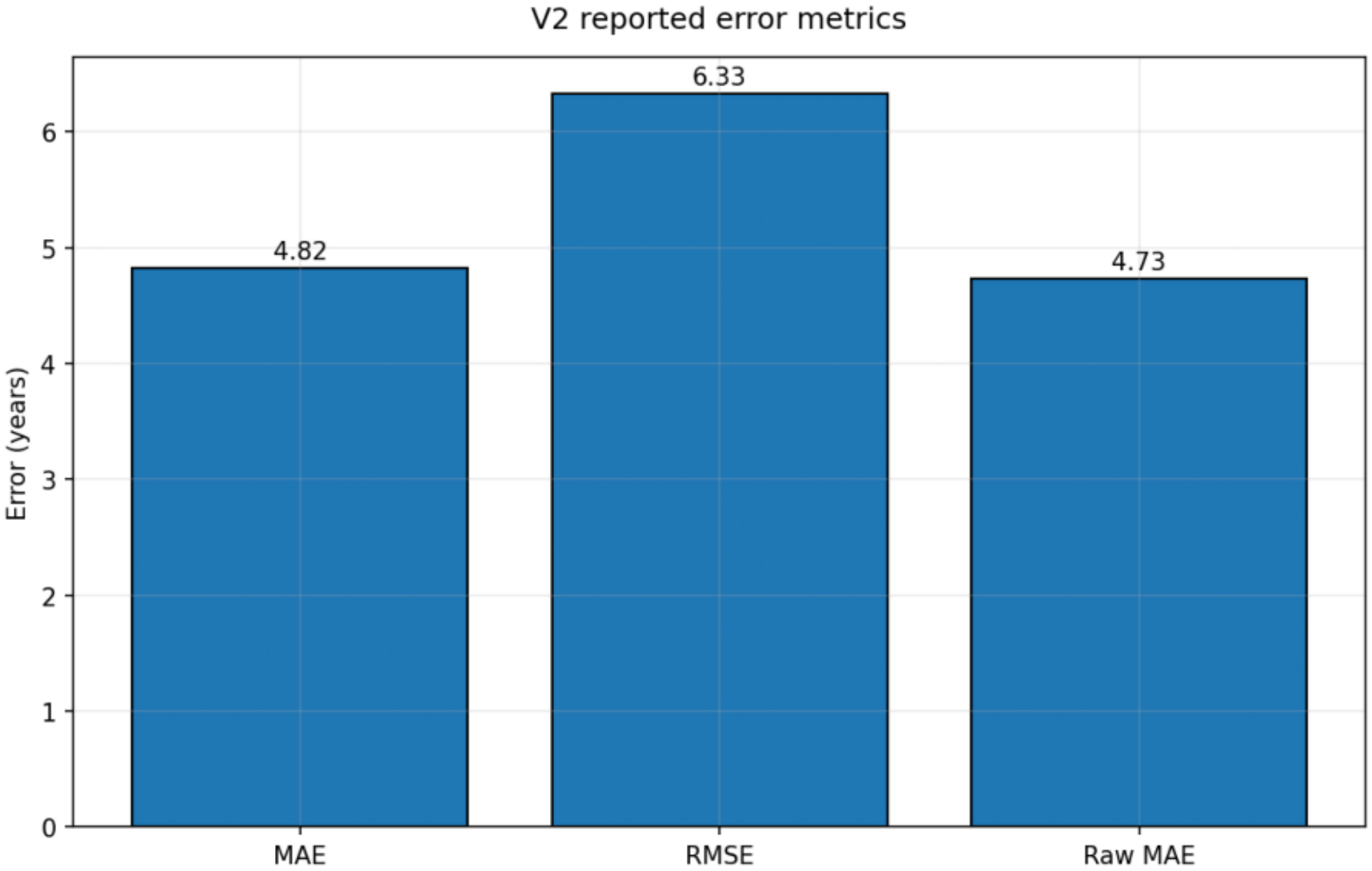

Absolute error by prediction-uncertainty quartile
Absolute error (years)
0
5
10
15
20
25
(0.03069999999999998, 0.738]
(0.738, 1.596]
(1.596, 2.525]
(2.525, 8.117]
Prediction SD quartile

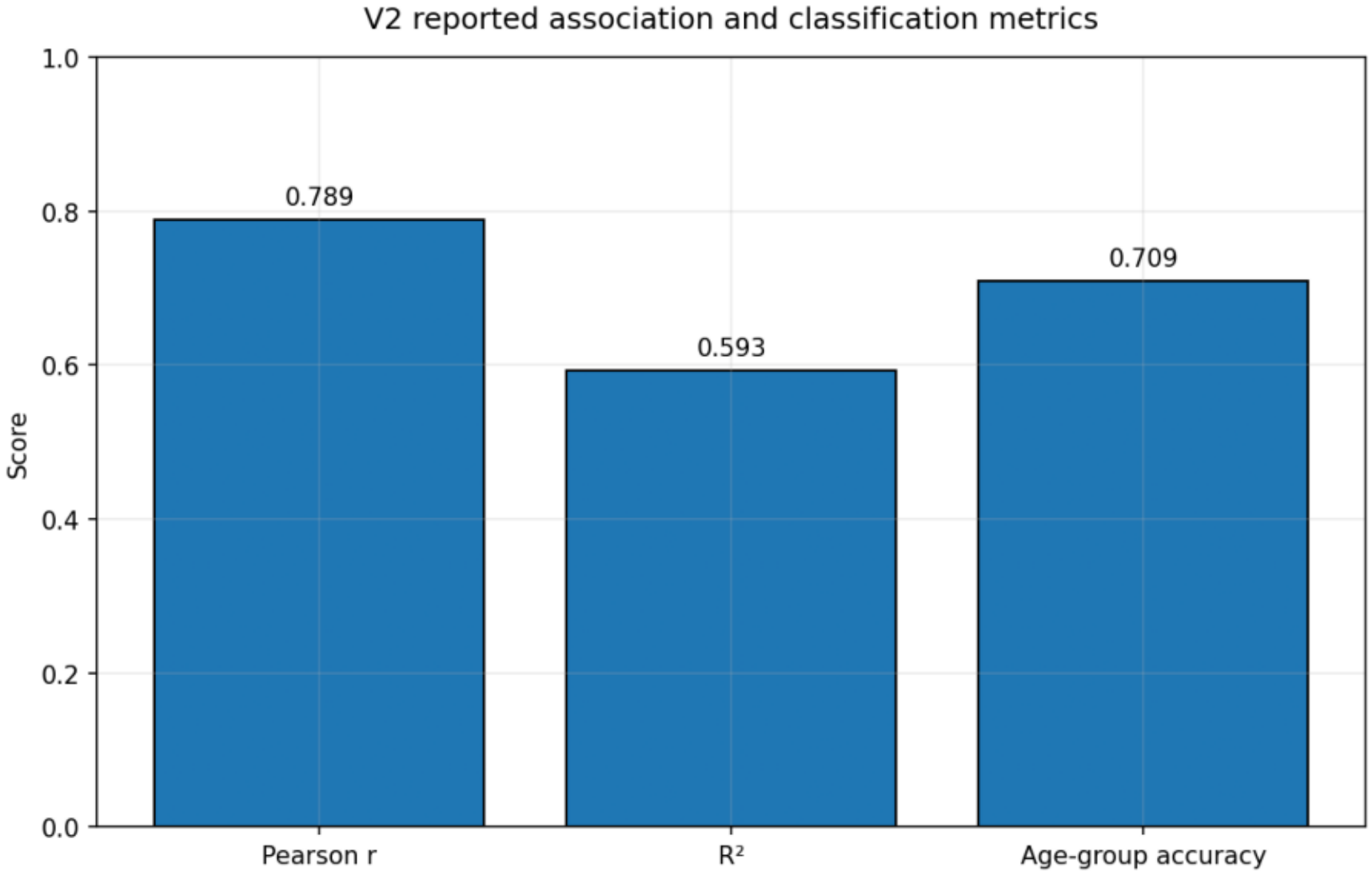

Repeated-run BAG: run 1 vs run 2
BAG run 2 (years)
BAG run 1 (years)
10
0
−10
−20
−20
−10
0
10

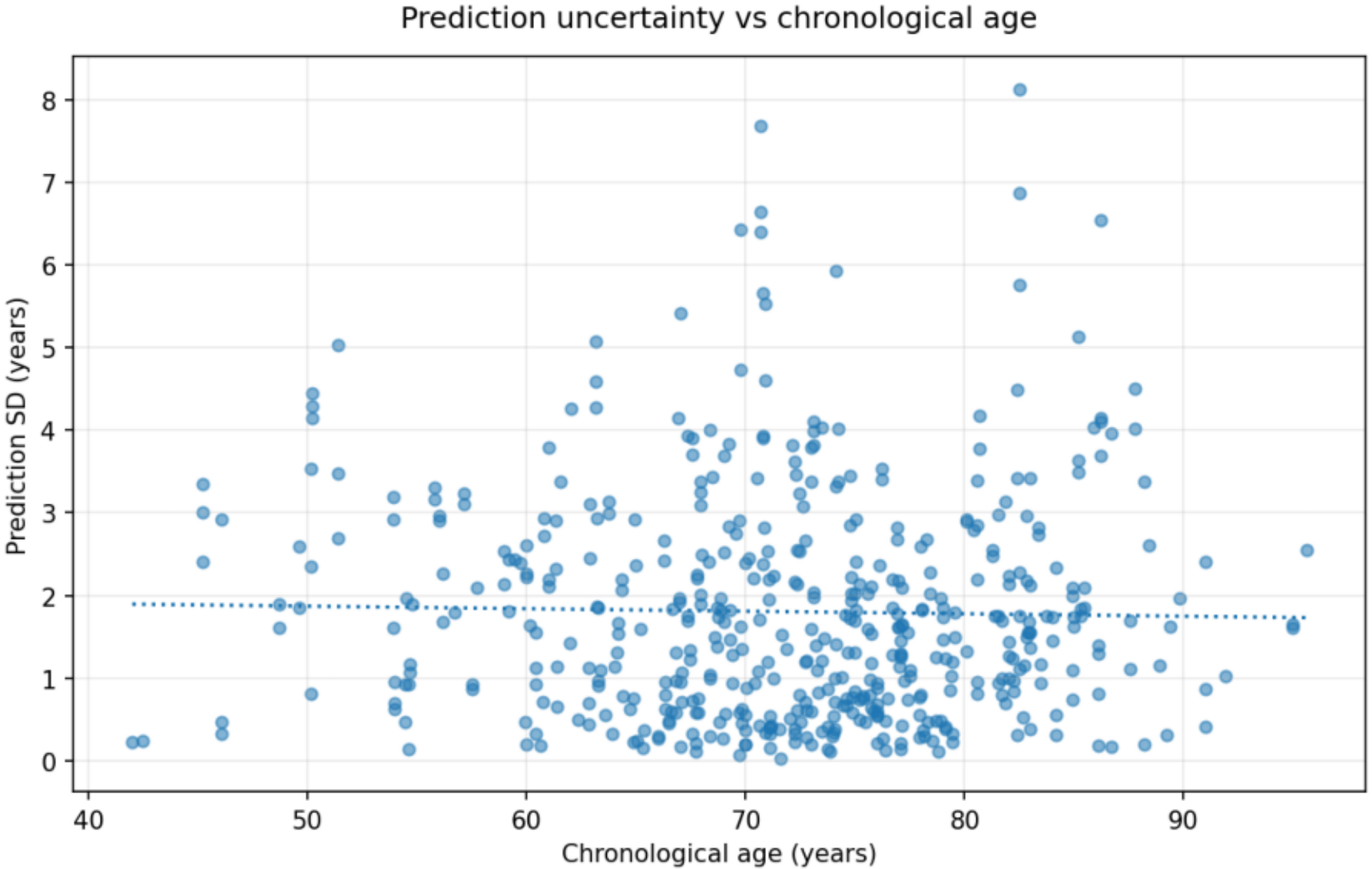

Repeated-run BAG difference distribution
50
40
30
20
10
0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
12.5

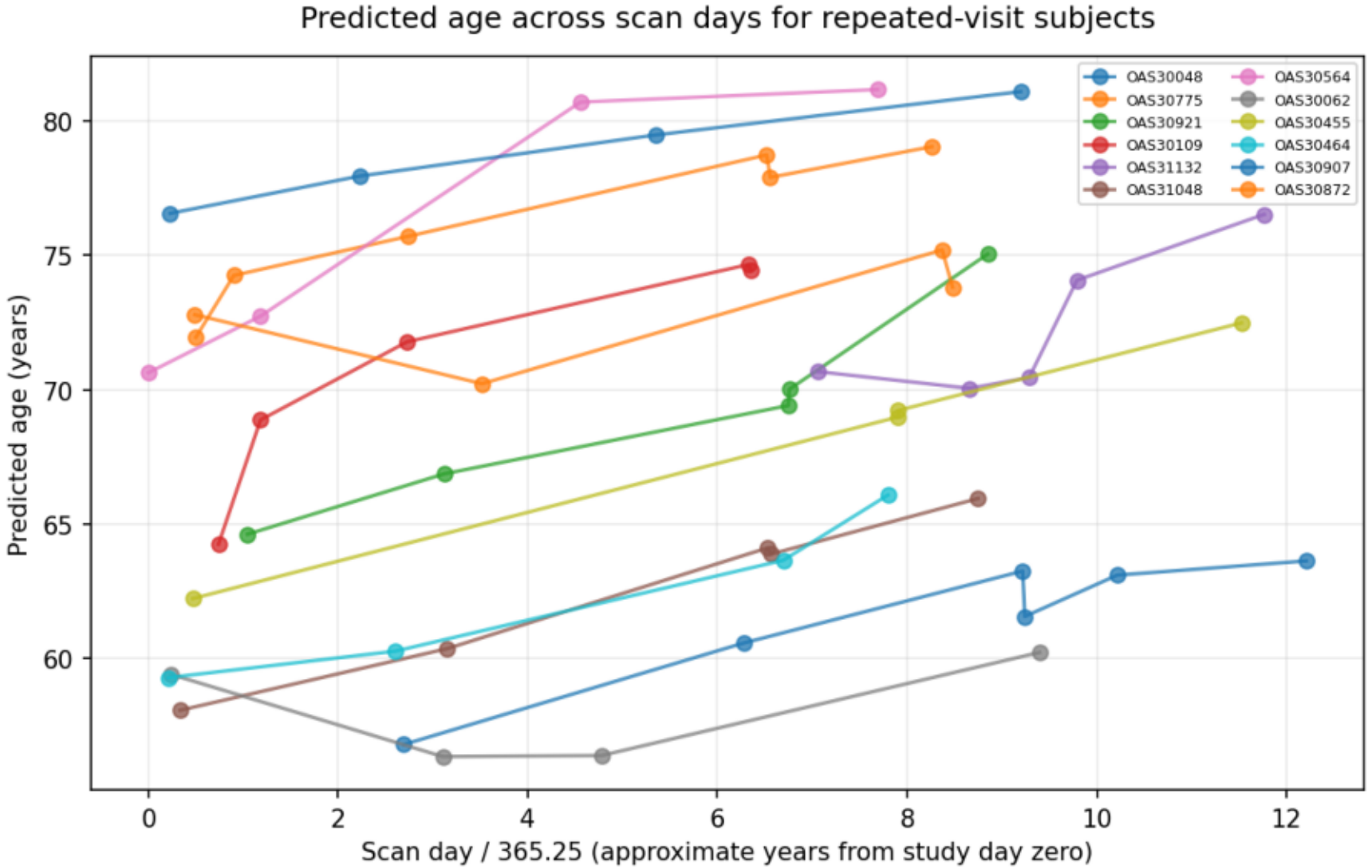

Repeat-run BAG agreement (Bland–Altman)
Run 2 − run 1 BAG (years)
Mean BAG across runs (years)
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−25
−20
−15
−10
−5
0
5
10
15

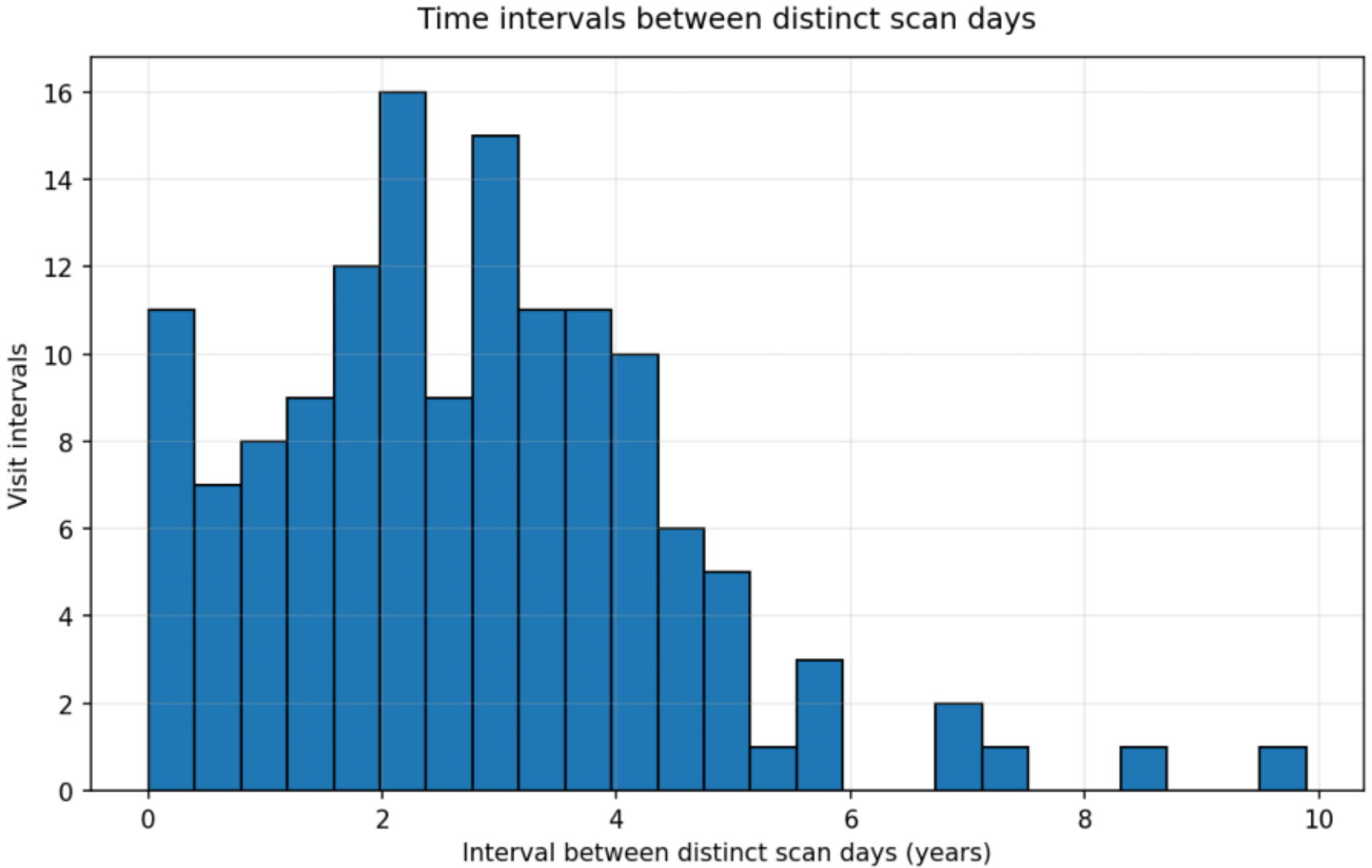

Repeated-run BAG difference by CDR
Run 2 − run 1 BAG (years)
12.5
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
0.0
0.5
1.0
CDR total score


V2 reported error metrics
Error (years)
0
1
2
3
4
5
6
4.82
6.33
4.73
MAE
RMSE
Raw MAE


V2 reported association and classification metrics
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.789
0.593
0.709
Pearson r
R²
Age-group accuracy


Prediction uncertainty vs chronological age
Prediction SD (years)
Chronological age (years)
0
1
2
3
4
5
6
7
8
40
50
60
70
80
90


Predicted age across scan days for repeated-visit subjects
Predicted age (years)
Scan day / 365.25 (approximate years from study day zero)
OAS30048
OAS30775
OAS30921
OAS30109
OAS31132
OAS31048
OAS30564
OAS30062
OAS30455
OAS30464
OAS30907
OAS30872


Time intervals between distinct scan days
Visit intervals
Interval between distinct scan days (years)
0
2
4
6
8
10
0
2
4
6
8
10
12
14
16